

Majority Voting in Sovereign Restructurings: The Aggregation Dilemma

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Abstract

Most sovereign bonds today feature “aggregated” collective action clauses (CACs), allowing the government to restructure *multiple* bond series at once through a vote of bondholders. A tradeoff arises between two aggregation procedures. We characterize how the cost-minimizing voting rule and restructuring offers depend on the heterogeneity within and across bonds. Furthermore, we analyze how the design of CACs affects investor bases, bond valuations, and vulture entry. Our results help rationalize the aggregation method employed by Argentina and Ecuador to restructure their international bonds in 2020, and shed critical light on the ongoing reform of Euro-Area CACs.

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Introduction

Collective action clauses (CACs) constitute a key pillar of the sovereign debt architecture and determine the fate of several trillion dollars of government debt securities worldwide, should they be subject to a restructuring. These contractual provisions enable the affirmative vote of a qualified majority of bondholders to bind a dissenting minority to the terms of a restructuring proposal put forth by the government—emulating the majority voting provisions of corporate insolvency laws.¹ For over two decades, CACs have been systematically inserted in the international bonds of major emerging market sovereign issuers with a view to alleviating the “holdout” problem in sovereign debt workouts, whereby a few creditors can jeopardize a restructuring and increase the cost of debt distress.² Since 2013, Euro-Area government bonds also incorporate majority voting provisions.

This paper is concerned with the different voting rules that a government can employ to restructure *multiple* bond series under CACs. While first-generation CACs operated through separate votes in each bond series, most sovereign bonds today include clauses that make the restructuring outcome depend (at least partly) on the *aggregate* vote across all series. In particular, under the so-called “enhanced” formulation of CACs introduced in 2014 and widely adopted since by emerging market issuers, the sovereign can choose—at the time it seeks approval for a restructuring of its bonds—between two aggregated voting procedures.³ Under the “two-limb” procedure, the government’s proposal goes through and affects all bond series if the consent share *across* series is above some aggregate threshold (typically 66.7%) and if the approval rate *within* each series is sufficiently high (typically above 50%). Instead, the “single-limb” procedure only relies on the aggregate vote—with a supermajority threshold typically set at 75%—but comes with the additional constraint, known as *uniform applicability* condition, that all bond series should receive the same restructuring terms (see [ICMA \(2014\)](#)).

Despite their widespread adoption, the workings and efficacy of enhanced CACs are still not well understood. While two-limb CACs have been in use since 2003, the addition of the single-limb procedure constituted the real innovation in the drafting of the enhanced ICMA version. The presumption in policy circles was that this procedure would become the method of choice to conduct distressed bond exchanges.⁴ This view was severely challenged when

¹In 2002, IMF’s First Deputy Managing Director Anne Krueger envisioned a statutory bankruptcy regime for countries, but the idea of a “sovereign debt restructuring mechanism” failed to gather political support.

²For further background information on sovereign CACs, see among others [Eichengreen and Portes \(1995\)](#), [Buchheit and Gulati \(2002\)](#), [IMF \(2014\)](#), and [Gelpern et al. \(2016\)](#). The coordination problem among creditors was first analyzed in the context of corporate workouts by [Roe \(1987\)](#) and [Gertner and Scharfstein \(1991\)](#).

³Since their introduction under the auspices of the International Capital Market Association (ICMA) and the International Monetary Fund (IMF), enhanced CACs have been inserted in 91% of new international sovereign bonds issued by emerging markets and low-income countries ([IMF \(2020\)](#)).

⁴For a rendition of that view, see [IMF \(2014\)](#) and [Sobel \(2016\)](#). In hindsight, [de la Cruz and Lagos \(2021\)](#) reflect: “The ICMA CACs included an innovation expected to enhance the ability of a sovereign to restructure its outstanding bonds [...]. The single-limb, uniformly applicable, modification was the star of the day.”

enhanced CACs were tested for the first time in the summer of 2020, with the restructuring of their international bonds by Argentina and Ecuador. While both governments had the option to use single limb, they both opted for *two-limb* aggregation, taking many commentators by surprise.⁵ In the Euro Area, nonetheless, whereas all public debt securities issued after January 2013 are subject to a two-limb version of CACs, member governments recently agreed to equip their future bond issues with CACs that *only* allow for *single-limb* aggregation.⁶

Motivated by their ubiquity and by these puzzling developments, this paper offers the first economic analysis of aggregated CACs. Taking the formulation of these standard provisions as given, we address the following set of questions: In a restructuring, which voting procedure allows the government to obtain the best possible deal? Can one rationalize the choice made by Argentina and Ecuador? What difference would the ongoing reform of Euro CACs make to the holdout threat and to restructuring outcomes across bonds? And more broadly, how does the design of CACs—and the anticipation of their optimal use by the government—affect allocations and prices in the bond market ahead of a restructuring? To analyze these issues, we adopt a framework that features multiple bond series and heterogeneous bondholders, allowing for heterogeneity within and across series. Whereas the former type of heterogeneity may arise from differences in discount rates, balance sheets, regulation, information, or litigation skills, the latter can be due to differences across bonds in terms of maturity or coupon rate, as well as in their investor bases.⁷

The paper makes four main contributions. First, we establish that, contrary to the conventional view, single-limb aggregation is not always optimal, and we clarify the circumstances under which each voting procedure dominates. Second, we provide detailed case studies documenting the concrete use of aggregated CACs in sovereign restructurings and apply the theory to these cases. Third, we show that the *option* to resort to single-limb voting, although it may not be used, deters the entry of holdouts—thereby providing a theoretical underpinning for the design of enhanced CACs. Fourth, we propose a dynamic equilibrium model of the sovereign bond market under aggregated CACs that can quantitatively account for the recovery rates and pre-restructuring valuations observed in the Argentine debt workout across maturities, and we use the calibrated model to assess the effects of alternative CAC designs.

Section 1 formulates the problem of a sovereign that seeks to restructure multiple bonds under enhanced CACs at minimal cost. When deciding whether to accept a restructuring

⁵See, e.g., [Setser \(2020\)](#) (“Ironically, Argentina is not actually using the most innovative feature of the new ICMA collective action clauses—so called single limb voting”) and [de la Cruz and Lagos \(2021\)](#) (“Argentina chose not to use ICMA’s prodigious child, the single-limb, uniformly applicable, aggregation method”).

⁶The decision, taken in November 2021 as part of a reform of the Treaty establishing the European Stability Mechanism, will enter into force once the parliaments of all member states have ratified the Amending Agreement. At the time of writing, Italy is the only member state that has not yet ratified the agreement.

⁷As put by [Setser \(2020\)](#) in the context of the 2020 Argentine restructuring: “Technically, all of the international sovereign bonds are of equal legal rank. But that does not mean all the bondholders want the same deal. Some think their bonds have a claim to be treated better than other equally ranked bonds.”

offer, bondholders have heterogeneous reservation values. To start with, we take the bond-specific reservation value distributions as given, and assume that all investors are atomistic. The choice of restructuring method by the sovereign involves a clear tradeoff: compared with the two-limb procedure, single-limb aggregation comes with a benefit arising from the removal of per-series constraints, but also with a cost as it prevents differentiated offers across bonds.

Technically, the government’s *aggregation dilemma* boils down to a comparison between two non-linear constrained optimization problems. While this comparison is generally non-trivial, a simple sufficient condition ensures that two-limb aggregation minimizes the total restructuring payout: two-limb voting dominates when none of the per-series constraints that it involves are binding. Indeed, the unique advantage of single-limb aggregation is then worthless, and single-limb voting only comes at a cost due to the uniform applicability restriction.

Section 2 characterizes the determinants of the cost-minimizing aggregation procedure when there are two bonds to be restructured. We show that single-limb aggregation is optimal only under specific circumstances—namely, when the bonds are sufficiently heterogeneous and the more “demanding” bond (whose holders tend to have higher reservation values) is small in the restructuring pool. Only then can it be valuable for the government to use the single-limb procedure, extending a uniform offer that attracts a low consent share from the more demanding bond. Indeed, when there is instead little heterogeneity across bonds, or when the more demanding bond is relatively large, the need to meet the high aggregate consent threshold ensures that none of the per-series constraints is binding—hence, the two-limb procedure dominates in either circumstance.⁸ Somewhat counterintuitively, our analysis reveals that the heterogeneity across bonds must be sufficiently *large* for the single-limb procedure with uniform offer to dominate.

Section 3 illustrates these results in a parametric example, assuming exponential reservation value distributions. We derive closed-form expressions for optimal offers, consent shares, and total restructuring payout under each aggregation procedure. Then, comparing restructuring costs across procedures, we obtain a necessary and sufficient condition—expressed as a function of the heterogeneity across bonds, their relative size, and the voting thresholds—that fully maps out regions of the parameter space where each aggregation method dominates.

Section 4 confronts the theory to the data. We delve into the specific details of the first three instances of sovereign bond restructurings in which aggregated voting was actually used—starting with the 2012 Greek “CAC retrofit”—and map these cases into the theory. In particular, considering the particular circumstances of the 2020 Argentine and Ecuadorian workouts under enhanced CACs, we find that our results help account for the use of two-limb voting by both governments. In the case of the Argentine “Macri bonds”, there was substantial heterogeneity across series, but the more demanding ones comprised a large fraction of the

⁸When the single-limb threshold is strictly greater than the two-limb aggregate threshold, two-limb voting also dominates when the share of the more demanding bond approaches zero.

restructuring pool. In the case of Ecuador, there was little heterogeneity across bonds. In both configurations, the unique advantage of single-limb aggregation was worthless.

Section 5 takes a step back and analyzes how the design of CACs affects the potential entry of (non-atomistic) vulture funds that may be able to derail a restructuring by acquiring sufficiently large blocking positions. We show that, even though it may not be employed on the equilibrium path under enhanced CACs, the single-limb procedure plays an important role as an *off-equilibrium* deterrent to holdout attempts: vultures refrain from taking positions that would block the restructuring of a bond under two-limb voting because they anticipate that the government would then resort to single-limb aggregation to bypass the need to obtain majority approval in each bond. In that sense, the hype that surrounded the introduction of the single-limb procedure in 2014 was quite justified. Taken together, our results imply that single-limb and two-limb aggregation should be viewed as complementary tools in the restructuring arsenal: whereas the former discourages the entry of potential holdouts, the latter often turns out to be the optimal restructuring method once vultures have been kept at bay. The ongoing reform of Euro-Area CACs, which replaces two-limb with single-limb aggregation, fails to take advantage of this complementarity.

Finally, Section 6 performs a quantitative exercise based on a dynamic extension of the model in which the bond market is populated by investors with heterogeneous discount rates and bonds explicitly differ in their maturities and coupon rates. Investors sort into bonds in an initial trading stage and, in the event of a restructuring, the government optimally chooses the recovery values and aggregation method. In equilibrium, the government’s restructuring approach depends on the (endogenous) reservation value distributions and on relative face values in the same way as in the reduced-form setup.⁹ The model is sufficiently rich to quantitatively account for many features of the Argentine restructuring of Macri bonds under enhanced CACs, including differences in recovery values and pre-restructuring prices across maturities. Using the calibrated model, we then run counterfactuals to illustrate how the design of CACs affects restructuring and pre-restructuring equilibrium outcomes.

Related Literature. Legal considerations lie at the heart of the sovereign debt literature. The seminal paper by [Eaton and Gersovitz \(1981\)](#) puts at center stage the lack of enforceability stemming from the absence of an international bankruptcy court and from the legal doctrine of sovereign immunity.¹⁰ [Bolton and Jeanne \(2007\)](#) analyze how the weak contractual environment and lack of a bankruptcy regime affect the types of debt claims used in international borrowing and lending. Another important feature of the sovereign debt market—the absence of a seniority structure across debt claims—can be traced back to the common use

⁹However, the impact of a change in the bonds’ relative size on the optimal voting procedure and payouts is more subtle in this setting, as it also affects the equilibrium reservation value distributions.

¹⁰Although sovereigns are no longer immune from suits in key jurisdictions such as the U.S. and the U.K., they are still effectively immune from attachment attempts by judgment creditors.

of pari passu and negative pledge clauses in sovereign debt contracts, giving rise to the debt dilution problem analyzed in Bolton and Jeanne (2009), Chatterjee and Eyigungor (2015), and Hatchondo et al. (2016) among others. Our contribution consists in analyzing how some recent and widely adopted contractual innovations shape sovereign debt workouts and the sovereign bond market ahead of restructurings.

Following earlier work by Bulow and Rogoff (1989), a recent strand of the sovereign debt literature—e.g., Yue (2010), Benjamin and Wright (2013), Hatchondo et al. (2014), Dvorkin et al. (2021) and Arellano et al. (2023)—focuses on the restructuring process, studying its quantitative importance and examining the determinants of the level of haircuts and length of negotiations. In particular, Pitchford and Wright (2012) analyze how the type of settlement process affects delays in an environment where the government cannot commit to settling on worse terms with holdouts. Instead, we zoom in on the role of contractually specified voting procedures in determining restructuring payouts, and analyze their broader impact on the bond market equilibrium.

Existing theoretical work on majority voting provisions in sovereign bonds—see, e.g., Bi et al. (2016), Engelen and Lambsdorff (2009), and Haldane et al. (2005)—concentrates on *series-by-series* CACs.¹¹ Considering setups featuring one bond and a single set of creditors, these papers study how strategic interactions and restructuring outcomes are affected by the introduction of a supermajority rule in place of a unanimity requirement. Closer to our paper, Bond and Eraslan (2010) analyze the optimal choice of voting rule by the debtor government. By design, this prior work is silent on cross-bond heterogeneity and aggregation. Instead, we adopt a setting with multiple bond series to address issues specifically related to the new enhanced CAC standard and to the design of Euro CACs.¹² Throughout, our analysis is positive in nature: our objective is to elucidate the workings of aggregated CACs in a restructuring and delineate some of their *ex ante* effects.¹³

On the empirical front, while the literature on private debt workouts—such as, e.g., Cruces and Trebesch (2013)—mostly focuses on aggregate restructuring outcomes, Sturzenegger and Zettelmeyer (2006, 2008) and Zettelmeyer et al. (2013) document within-deal variation in haircuts for selected episodes, Fang et al. (2021) analyze the combined impact of CACs and haircuts on participation rates within restructuring episodes, and Asonuma et al. (2023) systematically explore the relationship between haircuts and maturity at the bond level. None of these studies investigates the impact of *aggregated* CACs on restructuring outcomes. Our

¹¹This original (non-aggregated) version of CACs has long been inserted in English-law governed sovereign issues and became commonly adopted under New York law in the early 2000s.

¹²Early analyses of CACs in a one-bond setting are still relevant in practice in situations where a single bond is being restructured, or when the bonds being restructured only feature old-style series-by-series CACs.

¹³A full-fledged analysis of the optimal design of voting provisions in sovereign restructurings is left for future work. As argued by Dooley (2000) and Shleifer (2003), and as also pointed out by Ghosal and Thampanishvong (2013) in their analysis of series-by-series CACs, a tradeoff may arise between *ex post* and *ex ante* efficiency.

contribution consists in documenting how enhanced CACs were used when they were first tested and in mapping these cases into the theory. As more evidence on the restructuring of bonds with aggregated CACs accumulates in the future, our analysis can provide guidance for further empirical work, especially with regard to the choice of aggregation procedure.

Finally, our analysis is connected to a series of empirical studies that investigate how the inclusion of CACs affects sovereign bond prices, starting with the early contribution by [Eichengreen and Mody \(2004\)](#). Among recent work on the topic, [Picarelli et al. \(2019\)](#) and [Carletti et al. \(2021\)](#) focus on the pricing of Euro CACs, while [Chung and Papaioannou \(2021\)](#) document the impact of enhanced CACs in normal times and during distress episodes. [Donaldson et al. \(2022\)](#) examine more broadly how sovereign bond prices are affected by legal changes that modify the ease of restructuring. To our knowledge, no study has yet explored the differential price impact of various forms of aggregated voting provisions, and how this impact may vary across bonds. Our framework could be used to guide such investigations.

Outline. The paper proceeds as follows. Section 1 describes the tradeoff that the government faces in choosing its restructuring approach under enhanced CACs and formulates general sufficient optimality conditions. Section 2 offers an analytical characterization of the optimal aggregation procedure and restructuring proposal in the two-bond case, while Section 3 gives closed-form results and illustrations in a parametric example. Section 4 provides detailed information on the concrete use of aggregated CACs and discusses the relevant cases in light of the theory. Section 5 analyzes how the design of CACs affects the potential entry of vulture funds. Section 6 assesses the equilibrium impact of CACs in a dynamic extension of the model where bond-specific reservation value distributions arise endogenously from the sorting of investors into bonds. Section 7 concludes. All proofs are in the appendix.

1 The Restructuring Problem

This section lays out a general framework to analyze the government’s optimal use of aggregated voting provisions in the restructuring of multiple bonds, allowing for creditor heterogeneity both within and across bond series.¹⁴

Bonds, Restructuring Proposal, and Creditor Heterogeneity. There is a countable set $\mathcal{B}$ of bond series to be restructured, with $|\mathcal{B}| \geq 2$. The bonds are unsecured obligations and are of equal legal rank. The relative size of bond series i , expressed as a fraction of the face value of the entire restructuring pool, is given by λ_i , with $\sum_{i \in \mathcal{B}} \lambda_i = 1$. A restructuring proposal $\mathbf{w} = \{w_i\}_{i \in \mathcal{B}}$ made by the government consists of series-specific recovery values w_i per unit of face value. Upon receiving an offer from the government, a bondholder accepts

¹⁴In practice, a bond series typically corresponds to a unique ISIN, although it may sometimes include bonds with different ISINs but same payment terms.

if the proposed recovery value is at least as high as her own idiosyncratic reservation value, which we assume to be independent of the participation decisions of other investors.¹⁵

To capture within-series creditor heterogeneity as well as cross-series heterogeneity arising from differences in creditor base and/or bond characteristics (e.g., payment terms), we assume that the reservation values of holders of bond series i are distributed according to the cumulative distribution function F_i , known by the government.¹⁶ The share of holders of series i that consent to an offer w is thus equal to $F_i(w)$. For expositional simplicity, we initially assume that all investors are atomistic and take the CDFs as given. Section 5 analyzes an extension where a large investor can take blocking positions, while Section 6 provides a microfoundation where reservation values are both investor- and bond-specific.

Aggregated Voting Rules. Under enhanced CACs, the sovereign can choose among two voting procedures (or “modification methods”) at the time of the restructuring.¹⁷ According to the two-limb procedure, all bond series are restructured if the weighted consent share over the entire pool is greater than or equal to the aggregate threshold τ_2^a , and if the consent share within each series is no smaller than $\tau_2^s < \tau_2^a$. Instead, under single-limb voting, the uniform applicability condition requires that the same offer per unit of face value be made to all bond series, and CACs are triggered as long as the aggregate threshold τ_1 is reached.¹⁸ Uniform applicability is meant to ensure inter-creditor equity by avoiding that holders of large bond series dictate terms that are discriminatory against smaller series (see IMF (2014)).

1.1 Government’s Problem

The government seeks to restructure all bond series at minimal cost. The set of constraints that need to be satisfied by the restructuring proposal to achieve this objective depends on the elected modification method. Under two-limb voting, the restructuring offer $\mathbf{w} = \{w_i\}_{i \in \mathcal{B}}$ must satisfy the aggregate constraint

$$\sum_{i \in \mathcal{B}} \lambda_i F_i(w_i) \geq \tau_2^a, \quad (1)$$

¹⁵For an analysis of the strategic interactions that can give rise to the classical holdout problem in sovereign restructurings and the role of CACs in this context, see Bi et al. (2016) and Engelen and Lambsdorff (2009). While Schumacher et al. (2021) document the incidence of sovereign debt litigation, little evidence exists on holdout payoffs and their dependence on exchange participation. Cruces and Samples (2016) discuss the Argentine settlement with holdouts following the 2001 default.

¹⁶In practice, the government learns about reservation values during preliminary talks with bondholder committees. In Appendix B, we consider an environment where the government faces some uncertainty over the consent shares that a restructuring proposal will attract. We show that the presence of uncertainty does not alter the main insights from our analysis.

¹⁷Instead, the version of Euro CACs in place since 2013 only provides for two-limb aggregation, while the envisaged Euro-Area reform would *replace* two-limb voting with the single-limb procedure.

¹⁸Details on what “uniformly applicable” exactly means in practice are provided in Appendix D.

along with the per-series consent requirements

$$F_i(w_i) \geq \tau_2^s \quad \text{for all } i \in \mathcal{B}. \quad (2)$$

Instead, under the single-limb procedure, the “uniform” offer w must be such that

$$\sum_{i \in \mathcal{B}} \lambda_i F_i(w) \geq \tau_1.$$

In choosing the modification method and the offer w_i made to each bond series $i \in \mathcal{B}$, the government wishes to minimize the total payout

$$C = \sum_{i \in \mathcal{B}} \lambda_i w_i. \quad (3)$$

We proceed under the realistic assumption that $1/2 \leq \tau_2^s < \tau_2^a \leq \tau_1 < 1$.¹⁹

1.2 Optimal Restructuring Offers

Under single-limb voting, the optimal uniform offer u^* is such that the average consent share, weighted by the bond face values, is equal to the aggregate threshold τ_1 , i.e.,

$$\sum_{i \in \mathcal{B}} \lambda_i F_i(u^*) = \tau_1, \quad (4)$$

and the minimum restructuring cost, denoted by C_1 , is equal to u^* .

Instead, the optimal two-limb offer is the solution to the constrained problem

$$C_2 \equiv \min_{\{w_i\}} \sum_{i \in \mathcal{B}} \lambda_i w_i$$

subject to

$$\sum_{i \in \mathcal{B}} \lambda_i F_i(w_i) = \tau_2^a, \quad \text{and} \quad F_i(w_i) \geq \tau_2^s, \quad i \in \mathcal{B}.$$

Since $\tau_2^a > \tau_2^s$, the aggregate constraint (1) is binding and holds as an equality. However, it is a priori unclear which of the per-series constraints, if any, may be binding.

¹⁹In the ICMA template, the thresholds are $\tau_1 = 3/4$, $\tau_2^s = 1/2$, and $\tau_2^a = 2/3$. The ICMA standard also allows for series-by-series voting with a 75% threshold, but this method is dominated by two-limb voting.

Auxiliary Problem. It will be useful in our analysis to consider the auxiliary problem

$$\min_{\{w_i\}} \sum_{i \in \mathcal{B}} \lambda_i w_i \quad \text{subject to} \quad \sum_{i \in \mathcal{B}} \lambda_i F_i(w_i) = \tau, \quad (5)$$

for a given aggregate threshold τ . The government’s cost-minimization problem under single-limb and two-limb voting can be construed by reference to this problem. Under single-limb voting, the aggregate threshold is τ_1 and the offer needs to satisfy the additional “uniform applicability” restriction, which simplifies the problem into (4). Instead, under two-limb voting, the aggregate threshold is τ_2^a and the government needs to take into account the additional series-by-series constraints (2). Whenever the solution to the auxiliary problem for $\tau = \tau_2^a$ satisfies the latter constraints, it therefore coincides with the optimal two-limb offer.

Assuming differentiability, the auxiliary solution must satisfy the first-order condition

$$F'_i(w_i) = \xi^{-1}, \quad \text{for all } i \in \mathcal{B}, \quad (6)$$

where ξ denotes the Lagrange multiplier on the aggregate constraint. A marginal increase dw in the offer to series i raises the aggregate consent share by $\lambda_i F'_i(w_i) dw$ at a cost of $\lambda_i dw$. Condition (6) requires that the “bang for the buck” $F'_i(w_i)$ be equalized across all bonds.

1.3 The Aggregation Dilemma: Key Tradeoff

Comparing across procedures, the unique appeal of single-limb voting comes from the fact that it removes the need to satisfy the per-series constraints. However, resorting to this method does entail a cost arising from the uniform applicability restriction—let alone the higher aggregate consent threshold when $\tau_1 > \tau_2^a$. These observations immediately deliver sufficient conditions under which two-limb aggregation is optimal.

Proposition 1. *Two-limb aggregation is (at least weakly) optimal if one of the following conditions holds:*

- (i) *The optimal single-limb uniform offer, given by the unique solution u^* to (4), is such that $F_i(u^*) \geq \tau_2^s$ for all $i \in \mathcal{B}$;*
- (ii) *The solution $\hat{\mathbf{w}} = \{\hat{w}_i\}$ to the auxiliary problem (5) for $\tau = \tau_2^a$ is such that $F_i(\hat{w}_i) \geq \tau_2^s$ for all $i \in \mathcal{B}$.*

Furthermore, if condition (ii) holds, the optimal two-limb offer coincides with $\hat{\mathbf{w}}$.

Indeed, the unique advantage of single-limb aggregation is worthless if any of the two conditions holds. Hence, two-limb voting dominates in these configurations. Going beyond these general statements requires making further assumptions on the environment—that is, on the

number of bonds and their relative sizes λ_i , on the reservation value distributions F_i , and on the various voting thresholds.

2 Two-Bond Case

We now focus on the case where there are two bond series outstanding, H and L , with relative weights $\lambda_H = \lambda \in (0, 1)$ and $\lambda_L = 1 - \lambda$. We denote by $F_i : \mathbb{R}^+ \rightarrow [0, 1]$ the cumulative distribution function of reservation values for bond $i \in \{H, L\}$, which we assume to be strictly increasing and twice differentiable on $\mathbb{R}^+$ with $F_i(0) = 0$, and we denote by f_i the corresponding density function. We assume that holders of bond H tend to have higher reservation values, so that

$$F_H(w) < F_L(w) \quad \text{for all } w > 0. \quad (7)$$

Because it takes a more generous offer to reach a certain approval rate for bond H than for bond L , we will sometimes loosely refer to bond H as the “expensive” bond.

2.1 Single-Limb Aggregation

As per (4), the cost-minimizing uniform offer under single-limb voting is given by $u^* = u(\lambda, \tau_1)$, where $u(\lambda, \tau)$ is implicitly defined as the unique solution to the equation

$$\lambda F_H(u) + (1 - \lambda) F_L(u) = \tau.$$

The following observations obtain immediately.

Lemma 1. *The consent shares under the optimal uniform offer u^* are such that*

$$F_H(u^*) < \tau_1 < F_L(u^*).$$

Moreover, u^ is strictly increasing in τ_1 and in the relative size λ of the expensive bond, with*

$$\lim_{\lambda \downarrow 0} u^* = F_L^{-1}(\tau_1) \quad \text{and} \quad \lim_{\lambda \uparrow 1} u^* = F_H^{-1}(\tau_1).$$

The optimal uniform offer attracts a consent share from bond H below τ_1 , which is compensated by an approval rate from bond L higher than τ_1 . Naturally, the government needs to make a more generous offer when the aggregate threshold to be reached (τ_1) is higher, and when the more demanding series constitutes a larger share (λ) of the restructuring pool. Next, we establish conditions under which $F_H(u^*) \geq \tau_2^s$, in which case the optimal single-limb offer satisfies both of the per-series constraints imposed under two-limb voting.

Lemma 2. *If $F_L^{-1}(\tau_1) \geq F_H^{-1}(\tau_2^s)$, the optimal uniform offer u^* is such that $F_H(u^*) \geq \tau_2^s$ for any $\lambda \in (0, 1)$. Otherwise, $F_H(u^*) \geq \tau_2^s$ if $\lambda \geq \lambda_\dagger$, where $\lambda_\dagger$ is such that $u(\lambda_\dagger, \tau_1) = F_H^{-1}(\tau_2^s)$.*

The optimal uniform offer $u^* = u(\lambda, \tau_1)$ is more likely to satisfy the per-series constraints for high values of τ_1 and low values of τ_2^s . Besides, since the offer u^* induces an average consent share τ_1 , greater homogeneity across bonds makes it more likely that $F_H(u^*)$ is close to τ_1 and thus greater than τ_2^s . Finally, a higher relative size λ of bond H , by increasing the amount of the uniform offer u^* , makes it more likely that the per-series constraint on this bond be satisfied—as captured by the second part of the lemma.

2.2 Two-Limb Aggregation

Under two-limb aggregation, the optimal offer is the solution to

$$\min_{w_H, w_L} \lambda w_H + (1 - \lambda) w_L$$

subject to

$$\lambda F_H(w_H) + (1 - \lambda) F_L(w_L) = \tau_2^a, \quad (8)$$

$$F_i(w_i) \geq \tau_2^s, \quad i = H, L. \quad (9)$$

Auxiliary Problem. To characterize the optimal offer under two-limb aggregation, we consider the corresponding auxiliary problem

$$\min_{w_H, w_L} \lambda w_H + (1 - \lambda) w_L \quad \text{subject to} \quad \lambda F_H(w_H) + (1 - \lambda) F_L(w_L) = \tau_2^a.$$

Substituting the constraint inside the objective, the problem can be stated more concisely as

$$\min_{w_L \in \mathcal{W}} \lambda g(w_L) + (1 - \lambda) w_L, \quad (10)$$

where $g(w_L)$ denotes the offer that the sovereign needs to extend to the holders of bond H to satisfy the aggregate constraint, given an offer w_L to bond L .²⁰ The function g is strictly decreasing, reflecting the fact that if a more generous offer is made to bond L —thus raising the consent share for this series—a less appealing offer can be made to the other bond. Under technical conditions ensuring that the problem is well-behaved (see Remark A-2), the auxiliary solution is pinned down by the first-order condition

$$f_L(w_L) = f_H(g(w_L)). \quad (11)$$

²⁰The explicit definitions of the function g and of the set $\mathcal{W} \equiv \mathcal{W}(\lambda, \tau_2^a)$ that restricts offers w_L to be such that the aggregate constraint (8) can be met are given in the appendix (Remark A-1).

Optimal Two-Limb Offer. The optimal two-limb offer coincides with the auxiliary solution $\hat{\mathbf{w}} = (\hat{w}_H, \hat{w}_L)$ as long as the latter satisfies the per-series constraints (9). The following lemma provides conditions such that each of these two constraints is satisfied.

Lemma 3. *Suppose that the auxiliary problem (10) is strictly convex with an interior solution. The auxiliary solution $\hat{\mathbf{w}}$ satisfies the per-series constraint on bond H when $\lambda \geq (1 - \tau_2^a)/(1 - \tau_2^s)$ or if the inequality (A.12) given in the Appendix holds. It satisfies the per-series constraint on bond L when $\lambda \leq (\tau_2^a - \tau_2^s)/(1 - \tau_2^s)$ or if inequality (A.11) holds.*

When a bond is large relative to the total restructuring pool, the need to meet the high aggregate threshold τ_2^a ensures that the per-series requirement is satisfied for this bond. Hence, the per-series constraint on bond H (resp. bond L) is satisfied when λ is sufficiently large (resp. small), as stated in Lemma 3. When these sufficient conditions fail, the lemma provides an alternative set of conditions—expressed in the appendix in terms of the primitives of the problem—ensuring that the offers $(\hat{w}_H, \hat{w}_L)$ implied by the optimality condition (11) satisfy the per-series constraints. When instead the auxiliary solution is such that $\hat{w}_i < F_i^{-1}(\tau_2^s)$, the per-series constraint on bond i is binding. The optimal two-limb offer then sets $w_i = F_i^{-1}(\tau_2^s)$ and lowers the recovery value on the other bond to satisfy the aggregate requirement (8).

2.3 Optimal Restructuring Procedure

Considering the optimal choice of aggregation method by the government, the following proposition provides sufficient conditions under which two-limb voting dominates.

Proposition 2. *If either the conditions stated in Lemma 2 or those provided in Lemma 3 hold, two-limb aggregation is optimal. Moreover, two-limb voting dominates for λ sufficiently small when $\tau_1 > \tau_2^a$.*

The conditions stated in Lemmas 2 and 3 guarantee that the optimal uniform offer and the auxiliary solution satisfy all per-series constraints, respectively. In either case, two-limb voting dominates, as the unique advantage of the single-limb procedure is worthless (Proposition 1). Our discussion of Lemma 2 suggests that two-limb voting is more likely to dominate for high values of τ_1 and low values of τ_2^s , for low levels of heterogeneity across bonds, and for a high relative size λ of the expensive bond. To understand the second part of Proposition 2, one should note that, when the relative size λ of bond H approaches zero, the restructuring cost under any aggregation method boils down to the recovery value that the government needs to extend to bond L to attract a consent share from this bond that meets the aggregate threshold. Therefore, when the single-limb procedure comes with a higher aggregate threshold ($\tau_1 > \tau_2^a$), the restructuring cost is lower under two-limb voting for λ close to zero.

The last result of this section exhibits sufficient conditions for single-limb optimality.

Proposition 3. *Suppose that $F_L^{-1}(\tau_1) < F_H^{-1}(\tau_2^s)$ and the densities f_H and f_L are strictly decreasing and intersect only once, at a point $\tilde{w} \in (F_L^{-1}(\tau_1), F_H^{-1}(\tau_2^s))$. Then single-limb voting is optimal (i) for λ sufficiently small when $\tau_2^a = \tau_1$, and (ii) in the neighborhood of the point $\tilde{\lambda} > 0$ such that $u(\tilde{\lambda}, \tau_1) = \tilde{w}$ when $\tau_1 - \tau_2^a$ is not too large.*

As a pre-requisite for the premise of Proposition 3 to be satisfied, the difference $\tau_1 - \tau_2^s$ must be sufficiently small, and the heterogeneity across bonds must be large enough. Part (i) of the proposition then states that, when the more demanding bond is relatively small, single-limb voting is optimal in case the two procedures have the same aggregate thresholds ($\tau_1 = \tau_2^a$).²¹ Combining this result with the second part of Proposition 2 thus reveals that, when the more demanding bond is relatively small, the optimal voting procedure depends crucially on whether $\tau_1 > \tau_2^a$ or $\tau_1 = \tau_2^a$. Part (ii) of the proposition points to a parameter configuration in which the auxiliary solution entails making the same offer $\tilde{w} < F_H^{-1}(\tau_2^s)$ to the two bonds,²² thereby violating the per-series constraint for bond H . Single-limb voting is clearly optimal in such a configuration, as the uniform applicability restriction comes at no cost while the unique advantage of single-limb aggregation is valuable.

3 Parametric Example

We now consider a particular specification of the two-bond case where reservation values for each bond are exponentially distributed, that is,

$$F_i(w) = 1 - e^{-\frac{w}{\phi_i}}, \quad \text{for } w \geq 0. \quad (12)$$

As before, we suppose that holders of bond H tend to have higher holdout values. We thus proceed under the assumption that

$$\gamma \equiv \frac{\phi_H}{\phi_L} > 1.$$

3.1 Closed-Form Results

In the exponential specification, the consent shares under the auxiliary solution are given by²³

$$F_L(\hat{w}_L) = \frac{\lambda(\gamma - 1) + \tau_2^a}{\lambda(\gamma - 1) + 1} > \tau_2^a, \quad (13)$$

²¹When $\tau_1 = \tau_2^a$, the restructuring costs C_1 and C_2 coincide in the limit as λ goes to zero, but the assumptions of Proposition 3 ensure that C_2 grows faster than C_1 with λ in the neighborhood of zero (see proof in appendix).

²²By construction, the offer $\hat{\mathbf{w}} = (\tilde{w}, \tilde{w})$ satisfies the first-order condition (11) of the auxiliary problem. Moreover, for a relative weight $\tilde{\lambda}$ of bond H , it also satisfies the aggregate consent requirement (8) for $\tau_2^a = \tau_1$.

²³The analytical derivations for this section are provided in Appendix A.3.

and

$$F_H(\hat{w}_H) = \frac{\tau_2^a - (1 - \lambda)F_L(\hat{w}_L)}{\lambda} = \frac{1 + \lambda(\gamma - 1) - \gamma(1 - \tau_2^a)}{1 + \lambda(\gamma - 1)} < \tau_2^a. \quad (14)$$

Importantly, (13) and (14) reveal that only the individual constraint on bond H may ever be binding. The auxiliary consent shares (and corresponding offers $\hat{w}_L$ and $\hat{w}_H$) are increasing in the relative size λ of the more demanding bond. Moreover, one can check that $F_L(\hat{w}_L)$ is increasing in γ while $F_H(\hat{w}_H)$ is decreasing in γ —i.e., the spread in consent shares under the auxiliary solution is increasing in the degree of bond heterogeneity.

Optimal Two-Limb Offer. The optimal offers under two-limb aggregation coincide with the auxiliary solution as long as $F_H(\hat{w}_H) \geq \tau_2^s$. Since $F_H(\hat{w}_H)$ is increasing in λ and decreasing in γ , the inequality is more likely to hold for high values of λ and low values of γ . Indeed, one can show that $F_H(\hat{w}_H) \geq \tau_2^s$

- if $\lambda \geq (1 - \tau_2^a)/(1 - \tau_2^s)$, for all values of γ ;
- if $\gamma \leq (1 - \tau_2^s)/(1 - \tau_2^a)$, for all values of λ ;
- in the remainder of the parameter space for λ sufficiently large or γ sufficiently small.

Conversely, the consent requirement on the expensive bond H is binding when there is sufficient heterogeneity across the two bonds and bond H is relatively small.

Optimal Voting Procedure. When $F_H(\hat{w}_H) \geq \tau_2^s$, two-limb aggregation dominates since the per-series constraints have no bite. In the appendix, we show that the other sufficient condition for two-limb optimality—namely, that the optimal uniform offer be such that $F_H(u^*) \geq \tau_2^s$ —is also more likely to hold for high values of λ and low values of γ .

Proposition 3 provides us instead with a sufficient condition for single-limb optimality. We show in the appendix that the premise of the proposition is satisfied for sufficiently high values of γ . Then, single-limb voting is optimal for λ sufficiently close to zero when $\tau_1 = \tau_2^a$. If instead $\tau_1 > \tau_2^a$, as long as $\tau_1 - \tau_2^a$ is not too large, single-limb aggregation is optimal when the relative size λ of bond H is close to the value $\tilde{\lambda}$ —given by (A.29) in the appendix—such that the auxiliary solution would require giving the same recovery value to the two bonds, with a consent share for bond H below τ_2^s .

When none of the sufficient conditions holds, one can still compare the restructuring costs across the two aggregation procedures. Under two-limb voting, the constraint on bond H must then be binding, and the restructuring cost is given by

$$C_2 = \lambda \phi_H \log \left(\frac{1}{\zeta_2^s} \right) + (1 - \lambda) \phi_L \log \left(\frac{1 - \lambda}{\zeta_2^a - \lambda \zeta_2^s} \right),$$

where $\zeta_2^j \equiv 1 - \tau_2^j$ for $j \in \{s, a\}$. Defining

$$F(w) \equiv \lambda F_H(w) + (1 - \lambda) F_L(w) = 1 - \lambda e^{-w/\phi_H} - (1 - \lambda) e^{-w/\phi_L},$$

and noting that $u^* < C_2$ if and only if $F(u^*) = \tau_1 < F(C_2)$, we thus obtain a necessary and sufficient condition for single-limb optimality in terms of parameter values, namely:

$$\lambda (\zeta_2^s)^\lambda \left(\frac{\zeta_2^a - \lambda \zeta_2^s}{1 - \lambda} \right)^{\frac{1-\lambda}{\gamma}} + (1 - \lambda) (\zeta_2^s)^{\lambda\gamma} \left(\frac{\zeta_2^a - \lambda \zeta_2^s}{1 - \lambda} \right)^{1-\lambda} < 1 - \tau_1. \quad (15)$$

Below, we rely on this condition to represent regions of the parameter space in which each voting procedure dominates (see Figure 2).

3.2 Illustration

For the sake of illustration, we first assume that $\tau_1 = \tau_2^a = 2/3$ and $\tau_2^s = 1/2$, and we set the distributional parameters to $\phi_H = 0.7$ and $\phi_L = 0.2$ (i.e., the average reservation values for the two bonds are 70 and 20 cents on the dollar, respectively). Figure 1 depicts optimal offers (left panel) and consent shares (right panel) under the two aggregation methods as a function of the relative size λ of the expensive bond. Consider the two-limb modification method first. When λ is large enough, the auxiliary solution is feasible, both offers are strictly increasing

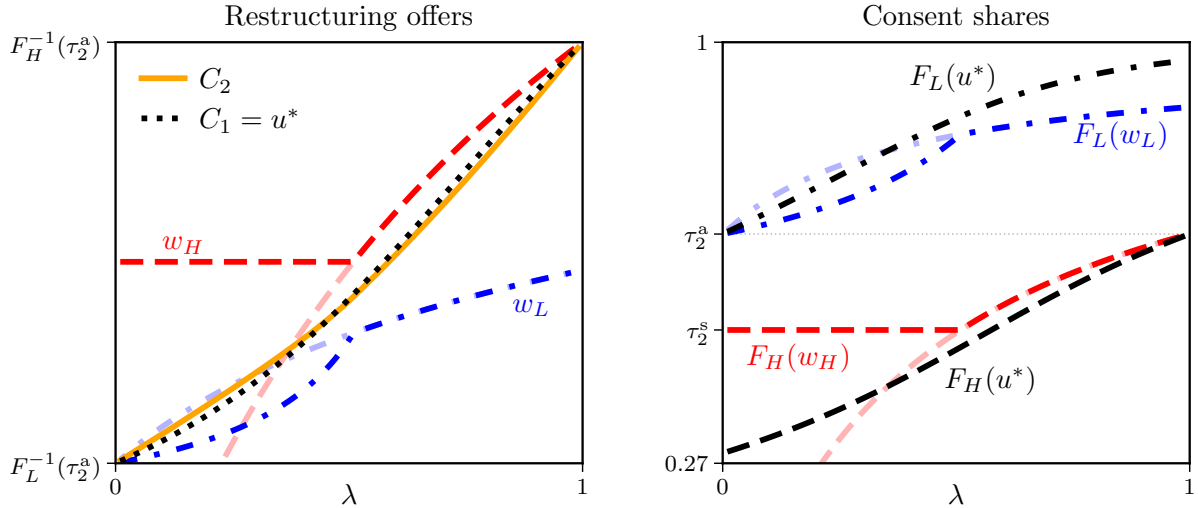


Figure 1: Optimal Restructuring Offers and Consent Shares.

Notes: Optimal offers and consent shares under two-limb voting are depicted in both panels as dashed red and dash-dotted blue lines for bonds H and L , respectively, with shaded lines corresponding to the auxiliary solution when it violates the constraint for bond H . The optimal uniform offer under single-limb aggregation is depicted as a dotted black line in the left panel, and the corresponding consent shares for bonds H and L appear as dashed and dash-dotted black lines in the right panel. The left panel also represents as a solid orange line the restructuring cost C_2 under two-limb voting. Parameter values are given in the main text.

in λ , and consent shares are larger than the series-by-series threshold τ_2^s . When instead λ is low, the series-by-series constraint for the expensive bond binds: w_H is flat, the consent share for bond H equals τ_2^s , and the government sets $w_L < \hat{w}_L$, so as to satisfy the aggregate consent requirement. Under single-limb voting, the optimal uniform offer and the associated consent shares are smoothly increasing in λ . The left panel of Figure 1 also displays the total restructuring cost as a function of λ for each of the two modification methods—showing that the two-limb method dominates for sufficiently high values of λ , while single-limb aggregation dominates when the share of the expensive bond is low.

Optimal Aggregation Method. Figure 2 represents the regions of the parameter space in which each of the two aggregation methods dominates, when $\tau_1 = \tau_2^a = 2/3$ (left panel) and when the thresholds are set as in the standard ICMA CACs, with $\tau_1 = 3/4$ and $\tau_2^a = 2/3$ (right panel). In both panels, the optimal restructuring method is determined according to the necessary and sufficient condition (15) as a function of the relative size λ of the expensive bond and the degree of heterogeneity across bonds, captured on the y-axis by $\log(\gamma) > 0$.²⁴

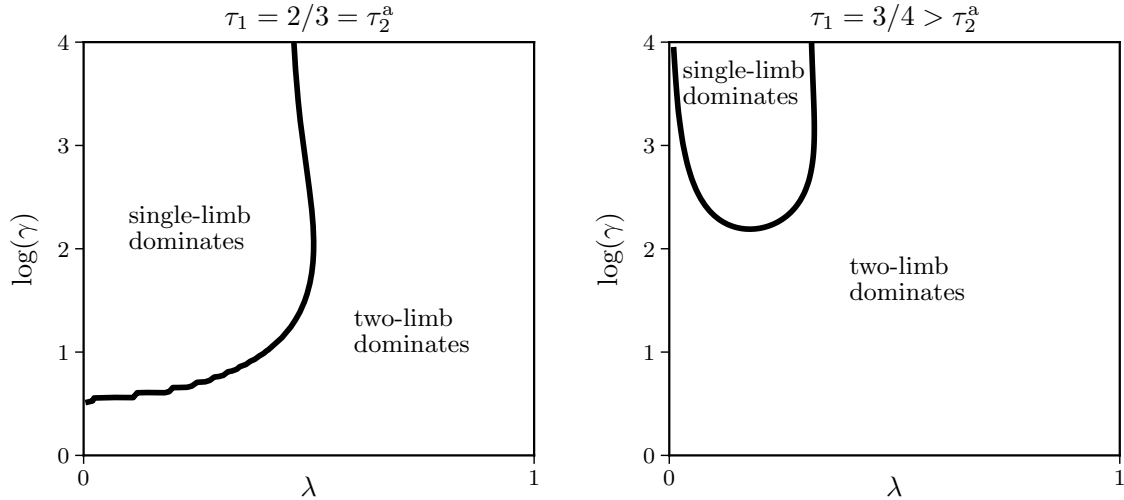


Figure 2: Optimal Voting Rule as a function of Relative Size (λ) and Bond Heterogeneity (γ).

Consistent with the rest of our analysis, the figure illustrates that single-limb aggregation only dominates when there is sufficient heterogeneity across bonds and the relative size of the expensive bond is not too large.²⁵ Naturally, the region where single-limb voting dominates shrinks when τ_1 increases. The right panel of the figure also illustrates that when $\tau_1 > \tau_2^a$, two-limb voting dominates in the neighborhood of $\lambda = 0$, in line with Proposition 2.

²⁴Figure A.1 in the appendix overlays the regions delineated by sufficient conditions for two-limb optimality. The necessary and sufficient condition (15) reveals some subtle non-linearities in the effects of λ and γ .

²⁵Figure B.4 in Appendix B generalizes this result to an environment with stochastic consent shares.

4 Aggregated CACs in Action: Case Studies

Turning to the data, this section provides detailed information on the use of aggregate voting in sovereign bond restructurings. We review the very first use of aggregation in the context of the Greek restructuring of 2012, and then document the approach taken by Argentina and Ecuador in the summer of 2020 to restructure their bonded debt under enhanced CACs. We discuss each of these cases in light of the insights derived in the previous sections. One obvious challenge in doing so is that we cannot directly observe the bond-specific reservation value distributions. However, in the cases of Argentina and Ecuador where the government opted for differentiated offers under two-limb aggregation, we identify the more “demanding” bonds as those that received a more generous offer, as suggested by the left panel of Figure 1. These bonds typically happen to be the ones that traded at a higher price ahead of the restructuring. We measure the relative size (λ) of bonds based on outstanding face values and assess the degree of heterogeneity across bonds (γ) based on their characteristics (i.e., differences in coupon rates and maturities) and pre-restructuring prices.

4.1 Greece 2012

The private debt workout conducted by the Greek authorities in March 2012 (euphemistically referred to as “Private Sector Involvement”, or PSI) is the largest such operation in history.²⁶ The restructuring pool consisted of 117 bonds, with a total face value of €205.6 billion. All bonds received the same exchange offer. Yet, the procedure employed to implement the restructuring differed across bonds according to their governing laws.

The Restructuring of Foreign-Law Bonds. About 10% of the public debt subject to the PSI consisted of bonds governed by foreign law. The vast majority of these instruments were governed by English law and featured *series-by-series* CACs, while a few other bonds did not incorporate any such provisions. In the latter, the exchange offer was opposed by a vast majority of bondholders. Among the former, CACs were triggered in about half of the series. In the other half, holdouts had acquired a blocking minority that prevented the activation of CACs—illustrating the limitations of series-by-series CACs as a defense against free riders.²⁷

Greek-Law Bonds and the Aggregated CAC Retrofit. The other 90% of the restructuring pool consisted of bonds governed by domestic law—which mostly consisted of Greek government bonds (GGBs) with a face value of €177 billion. None of these instruments originally incorporated CACs. However, since they were issued under local law, it only took a change of statute to restructure them. In February 2012, the Greek authorities passed the

²⁶This section borrows extensively from the analysis of the Greek PSI by Zettelmeyer et al. (2013).

²⁷The share of holdouts in some of these bonds was as high as 100%. Interestingly, holdouts did not block the activation of CACs in the *largest* series, such as a €5.5 billion bond maturing in April 2016.

Greek Bondholder Act to retroactively insert a majority voting provision in all local law GGBs. The retrofitted CAC only featured an *aggregate* consent requirement, with a supermajority threshold set at 66.7% of the face value taking part in the vote. It would have cost a wannabe holdout about €59 billion to secure a blocking position.

The GGBs subject to the CAC retrofit were quite heterogeneous. One week before the release of the exchange offer, shorter-maturity bonds traded about 50% higher than long-term bonds. Nonetheless, although it was not bound by any “uniform applicability” constraint, Greece offered the same exchange package to all bonds. Overall, the required majority was largely met and all local-law GGBs were entirely restructured, ensuring the large success of the PSI.²⁸

Discussion. The Greek PSI enabled the restructuring of nearly €200 billion of bonded debt, and the aggregated CAC retrofit was key to its success.²⁹ Nonetheless, one may wonder why Greece did not avail itself of the possibility of making differentiated offers across bonds.³⁰ According to Zettelmeyer et al. (2013), the main reason why Greece opted for a single offer was to streamline the deal as much as possible to complete it before the redemption of a €9.7 billion GGB due 20 March 2012. In theory, nothing prevents the solution to the auxiliary problem (5) from entailing a uniform offer across heterogeneous bonds. Another consideration that may be relevant in this case is that long-term bonds were primarily held by institutional investors who took part in negotiations with the authorities, whereas short-dated instruments had recently been bought by distressed debt funds at large discounts in the hope of a quick full repayment. The government’s cost objective (3) may have involved distorted weights reflecting the higher perceived cost of payouts to vultures.

4.2 Argentina 2020

The restructuring of Argentine international bonds conducted by the Fernández administration between April and August 2020 involved two distinct sets of bonds.³¹ On the one hand, the “Kirchner bonds”, comprising 8 series, had been issued as part of the 2005-2010 exchange

²⁸Holders of the more expensive short-term bonds may have been more reluctant to accept the uniform exchange offer. Unfortunately, instrument-level data on participation rates is not publicly available for GGBs.

²⁹It is unlikely that another Euro-Area government may be able to exploit its “local law advantage” in the same way as Greece did. According to Grund (2017), the jurisprudence of European courts in the wake of the Greek PSI sets important limits on future potential interference with bondholders’ property rights. Another important difference is that the original documentation of local-law GGBs did not specify any restructuring procedure, so the Greek Bondholder Act was actually filling a void, whereas Euro-Area government bonds now incorporate majority voting provisions.

³⁰Extending a single offer when attempting to restructure multiple sovereign bonds is not the norm. According to Asonuma et al. (2023), over the 1998-2019 period, 19 restructurings of international sovereign bonds (out of 27 in total) involved multiple bonds, 11 of which entailed differentiated offers across series.

³¹Appendix C.2 provides detailed information on eligible bonds, exchange offers, participation results, and recovery values for the Argentine restructuring, along with references to relevant data sources.

operations following Argentina’s 2001 default. On the other hand, the 17 series of “Macri bonds” had been issued between 2016-2019 under the Macri administration—once Argentina had finally settled with tenacious holdouts 15 years after its default and thereby regained access to the international bond market. The total face value of these eligible bonds amounted to about \$66 billion, of which the Macri bonds accounted for almost two thirds (\$42 billion).

While both sets of bonds had been issued under English or New York law, they contained two distinct versions of aggregated CACs. Specifically, the Kirchner bonds included a two-limb version of CACs, with an aggregate threshold of 85% and a 66.7% per-series threshold. Instead, Macri bonds featured enhanced ICMA CACs, leaving the sovereign the choice between two-limb aggregation (with aggregate and per-series thresholds 66.7% and 50%, respectively) and the “uniformly applicable” single-limb procedure, with a 75% aggregate threshold.

Each set of bonds was quite heterogeneous.³² Among the Kirchner bonds, the most substantial difference was between the “Discount” and “Par” bonds, with the former having lower duration and higher prices.³³ On the other hand, Macri bonds differed considerably in their maturities, which ranged from 2020 all the way to 2117. Since Argentine bonds were not accelerated following the payment default of May 2020, differences in maturities and coupon rates remained material at the time of the restructuring.

Two-Limb CACs—The Restructuring of Kirchner Bonds. In dealing with Kirchner bonds, Argentina had to employ a two-limb aggregation procedure.³⁴ The Republic made differentiated exchange proposals, extending a more generous offer to the holders of Discount Bonds, which were more valuable (see Tables C.4 and C.8). Wary of not meeting the 66.7% threshold in all series, Argentina controversially announced that, should one or more series fail to reach that threshold, it would allow itself to “redesignate” ex post the aggregation pool.³⁵ Holdouts did obtain blocking minorities in two of the Par Bonds (see Table C.6). But thanks to redesignation, CACs were triggered and the other six Kirchner bond series were entirely restructured according to the proposal.

Enhanced CACs—The Restructuring of Macri Bonds. In dealing with the Macri bonds, the Republic could choose between two-limb and single-limb aggregation. The government did select the two-limb procedure and made differentiated offers across series, extending a more attractive offer to the 2020-2036 maturities than to the 2046-2117 bonds (see Tables C.5 and C.9). The aggregate requirement was met, and all Macri series obtained more than 50% per-series approval, so that all of them were fully restructured (Table C.7).

³²Tables C.1 and C.2 in the appendix provide details on bond characteristics. [Setser \(2020\)](#) discusses the implications of bond heterogeneity for the best restructuring approach in the Argentine context.

³³The names “Discount” and “Par” refer to the terms of the conversion offer extended in 2005 and 2010.

³⁴The alternative would have been to use series-by-series voting with a 75% threshold. Given the aggregate threshold of 85% under two-limb voting, the choice of modification method was not trivial, but the lower per-series threshold (66.7%) made it more difficult for a potential holdout to acquire a blocking minority.

³⁵The analysis conducted in Appendix B in a setting with stochastic consent shares allows for redesignation.

Discussion. The presence of aggregated CACs made the 2020 Argentine debt workout much smoother than the 2005-2010 restructuring, with about \$63.9 billion of bonded debt successfully restructured in just a few months. The blocking of CACs in two of the Kirchner series, however, demonstrates that the two-limb voting procedure is still vulnerable to holdouts—a point that we further elaborate in Section 5.

Did Argentina pick the aggregation procedure optimally in dealing with its Macri bonds? The analytical insights derived in Sections 2 and 3 suggest so. There was clearly a lot of heterogeneity across Macri series, with very long-term bonds (including the famous “century bond” expiring in 2117) trading at higher discounts ahead of the restructuring.³⁶ However, the more expensive, shorter-maturity bonds accounted for about 3/4 of the face value of the Macri pool. Given the large weight of these more demanding series, a uniform offer would have had to attract consent from more than 50% of these bonds in order to obtain the required 75% average approval under single-limb aggregation.³⁷ Hence, the unique advantage of the single-limb procedure (i.e., removing the need to satisfy per-series constraints) was worthless. Mapped into Figure 2, the case of Argentine Macri bonds is one where two-limb voting dominates due to a high λ .³⁸

4.3 Ecuador 2020

The Ecuadorian restructuring of 2020 involved 10 bond series, with a total face value of \$17.4 billion. One of the bonds, issued in 2014, featured the old-style, within-series version of majority voting provisions. The other 9 bonds instead incorporated the aggregated ICMA model of CACs with standard threshold values (as in the Macri bonds). Further details on the eligible bonds can be found in Appendix C.3.

Like Argentina, Ecuador opted in favor of the two-limb procedure to restructure the bonds subject to enhanced CACs. According to the government’s proposal extended in July 2020, each instrument could be exchanged for a basket of new bonds—with the same overall reduction in face value. The composition of the exchange basket differed across series, with the duration of the exchange basket increasing in the maturity of the original instrument, but the exchange offers were overall quite similar across series (see Tables C.12 and C.13).³⁹ By

³⁶In early March 2020, shorter-maturity bonds traded at prices 12% higher than longer-maturity bonds.

³⁷Indeed, even assuming 100% consent among the longer-term bonds, it would have taken a 67% approval rate among the shorter-maturity bonds to reach the single-limb aggregate threshold.

³⁸An additional feature of the Macri-bond CACs—known as the “carry-over” clause—only reinforced the appeal of the two-limb procedure. This clause provided that, in assessing the aggregate criterion under the two-limb method, the Republic could decide to also take into account the votes of the Kirchner bonds. Assuming that the 85% aggregate threshold was met among the latter, given that Macri bonds accounted for about 2/3 of the face value of all eligible bonds, the clause effectively reduced the average consent share that had to be met across Macri bonds under two-limb voting to 57.5%.

³⁹At exit prices, the value of exchange baskets across eligible bonds ranged between 58 and 59.9 percent.

the end of August 2020, the restructuring was completed: the conditions for the activation of CACs had been met and all bond series were to be fully restructured.⁴⁰

Discussion. How does the Ecuadorian case map into theory? In this case, the degree of heterogeneity across the nine series subject to enhanced CACs was arguably mild. The range of maturities was indeed much smaller than in the case of Macri bonds. As a matter of fact, the approval rates induced by rather similar exchange offers were very homogeneous across bonds, varying within a range of 1 percentage point. With little heterogeneity across series, none of the per-series constraints is binding given the need to reach the higher aggregate threshold. Hence, two-limb aggregation is indeed optimal in such a (small- γ) configuration.

5 Aggregated CACs and Vultures

Taking one step back, this section analyzes how the design of CACs affects the potential entry of vulture funds who may be able to acquire large positions and block bond restructurings. Our analysis points to a potential complementarity between two-limb and single-limb aggregation that the reform of Euro-Area CACs would fail to exploit.

5.1 Setup with Vulture Entry

We consider an extension of the setup of Section 2 in which, prior to the restructuring episode, a vulture fund may potentially acquire bonds H and L from the continua of heterogeneous investors who hold them. We denote by ψ_i the share of bond i acquired by the fund, and we denote by q_i the price paid by the fund per unit of face value. We assume that the fund needs to pay a fixed cost ε_i to enter the market for bond i , which can be interpreted as the search cost of finding a counterparty in a highly decentralized marketplace.

Blocking Positions. In a restructuring, holding a sufficiently large position effectively allows the fund to prevent the triggering of CACs. We assume that the fund derives a gross expected payoff π_i per unit of face value that it holds from blocking the restructuring of bond i . In particular, single-limb aggregation can be blocked if $\sum_i \lambda_i \psi_i > 1 - \tau_1$, and the fund can block the restructuring of bond i under the two-limb procedure when $\psi_i > 1 - \tau_2^s$. We assume that the restructuring offer is subject to a high minimum participation threshold, implying that if CACs are not triggered for a bond, the bond is left entirely unrestructured.

Bond Market Prices. The price q_i at which the fund may acquire a position in bond i depends on the payoff that atomistic investors expect at the subsequent restructuring stage.

⁴⁰Investor heterogeneity in litigation proclivity was manifested in late July 2020, when two funds, Contrarian and GMO, initiated (short-lived) legal action against Ecuador in a New York district court.

If the market understands that the fund will be unable to prevent the restructuring of bond i , the price q_i coincides with the government's optimal restructuring offer w_i . If instead the fund's acquired position later enables it to block the restructuring of bond i , investors would ultimately be left with their own reservation values for this bond. In that case, each investor is thus willing to sell as long as the market price q_i is above her reservation value, implying that the fund can acquire a share ψ_i of bond i at a cost of $F_i^{-1}(\psi_i)$ per unit of face value.

Cost of Holdout Strategies. Since $\tau_2^a \leq \tau_1$, the minimum cost at which the fund can block the restructuring of *both* bonds under either aggregation method is given by

$$\kappa_2 \equiv \inf_{\{\psi_i\}} \sum_i \mathbf{1}_{\{\psi_i > 0\}} \left(\lambda_i \psi_i F_i^{-1}(\psi_i) + \varepsilon_i \right) \quad \text{s.t.} \quad \sum_i \lambda_i \psi_i > 1 - \tau_1.$$

Likewise, the minimum cost at which the fund may be able to prevent the restructuring of bond i via two-limb aggregation is

$$\kappa_i \equiv \lambda_i (1 - \tau_2^s) F_i^{-1}(1 - \tau_2^s) + \varepsilon_i, \quad i = H, L.$$

We assume, without loss of generality, that bond L is the cheapest to block, i.e., $\kappa_L < \kappa_H$.

Limited Financial Resources. We assume that the fund has limited resources relative to the size of the outstanding bond series, which restricts its potential ambitions as a holdout. Namely, we assume that the fund's resources e are such that $\kappa_L < e < \min\{\kappa_H, \kappa_2\}$. We thus focus on the case where the only holdout strategy that the fund may consider is to acquire a blocking position in one bond (namely, bond L) that would prevent the activation of CACs for this bond under two-limb aggregation.

5.2 Equilibrium

In equilibrium, the outcome of the entry stage in which the fund decides on (ψ_H, ψ_L) must be consistent with the outcome of the restructuring stage. It is straightforward to see that, under the previously stated assumptions, the fund would refrain entirely from entering the market for bond H . Indeed, since any position it may feasibly acquire in that bond would be non-blocking, the price q_H would be equal to the restructuring payout w_H , and the fixed entry cost would be enough to keep the fund at bay. By the same logic, any position $\psi_L > 0$ that is non-blocking is suboptimal. The only remaining question is: would the fund decide to acquire a blocking position in bond L or would it rather stay away altogether?

Two-Limb CACs. Consider first the case where CACs only allow for two-limb aggregation. If the fund contemplates acquiring a blocking position in bond L , the optimal blocking share

ψ_L^* is given by the solution to the constrained problem

$$B \equiv \sup_{\psi_L > 1 - \tau_2^s} \lambda_L \psi_L \left(\pi_L - F_L^{-1}(\psi_L) \right) - \varepsilon_L \quad \text{s.t.} \quad \lambda_L \psi_L F_L^{-1}(\psi_L) + \varepsilon_L \leq e.$$

As long as $B > 0$, it is optimal for the fund to purchase a share ψ_L^* of bond L and then block the restructuring of the bond in order to get the holdout payoff π_L , with net benefit B .

Enhanced CACs. Suppose now that the government can choose between the two aggregation methods. How would it respond if the fund builds a blocking position in bond L ? Under the two-limb method, bond L would have to be left unstructured. Instead, under single limb, the government would pick the uniform offer u^* that attracts an aggregate consent share τ_1 . As long as the cost of leaving one bond unstructured is sufficiently high,⁴¹ the government would optimally respond using single-limb aggregation, thus defeating any attempt to block bond L . Hence, the fund would optimally refrain from entering, and the government would choose its restructuring approach according to the logic outlined in Section 2.

5.3 Vulture Entry and the Design of CACs

The above analysis establishes that, although single-limb aggregation may not be used in equilibrium under enhanced CACs, it does serve as an *off-equilibrium* threat that deters potential holdouts from acquiring a blocking position in a bond. Indeed, in the equilibrium under enhanced CACs, the vulture fund optimally decides to stay at bay. Off equilibrium, the fund considers building a blocking position $\psi_L > 1 - \tau_2^s$ at a cost $q_L = u^*$ per unit of face value, where the market price q_L reflects the (correct) expectation that the government would defeat the holdout attempt using single-limb aggregation with uniform offer u^* —so that the fund would ultimately make a loss due to the fixed entry cost. Hence in the restructuring, the government ends up selecting the aggregation method according to the relative size of the bonds and the cross-bond heterogeneity in the reservation values of non-vulture investors, as illustrated in Section 3. Unless a relatively small bond is held by highly demanding creditors, the government would thus be using two-limb aggregation in equilibrium. However, if the option to use single-limb aggregation were to be removed, the equilibrium outcome would be dramatically different: the vulture fund would optimally decide to acquire a blocking position in one bond, making its restructuring impossible or more costly.

Discussion of Euro CACs. The Euro CAC standard in place since 2013 is a two-limb version of CACs. This version, as discussed above and as revealed by the Argentine restructuring of Kirchner bonds, is vulnerable to holdouts blocking a subset of bond series. In that respect, moving to the single-limb version, as contemplated under the ongoing Euro-Area reform, would

⁴¹ In practice, this cost may arise from not being able to re-access the international bond market.

constitute a more effective response to the holdout problem. However, by substituting one for the other, this reform fails to take advantage of the complementarity between the single-limb and two-limb procedures: while the former keeps vultures at bay, the latter can lower the restructuring cost by allowing for differentiated offers. For a government willing to obtain the smoothest and best restructuring deal, having the option to choose between the two methods would clearly be preferable.⁴²

6 Quantitative Application

In this section, we embed the government's restructuring problem in a dynamic setting where defaultable bonds explicitly differ in their maturities and coupon rates. In this framework, bond-specific creditor bases and reservation value distributions are determined endogenously, thus providing a microfoundation for the reduced-form formulation adopted in prior sections. We assess the model's quantitative performance against the 2020 Argentine restructuring data, and employ the calibrated model to illustrate how the design of CACs affects restructuring outcomes as well as the bond market equilibrium prior to restructuring.

6.1 Extended Setup

Time is continuous and the market is populated by a continuum of risk-neutral investors who differ in their discount rates r , distributed according to the cumulative distribution function G on the interval $\mathcal{R} = [r_{\min}, r_{\max}]$.

Bonds. There are two bonds S and L . The face value of bond $i \in \{S, L\}$ decays exponentially at rate δ_i . We shall assume that $\delta_S > \delta_L$, hence bonds S and L can be thought of as short-term and long-term bonds, respectively.⁴³ The government may default on its bonds, and the arrival time of default is exponentially distributed with parameter η . While there is no default, bond i pays out coupons continuously, at rate c_i . Upon occurrence of a default, bondholders receive a bond-specific recovery rate w_i per unit of face value, as further specified below. We assume that the government continuously issues new bonds, so that the relative face values of short- and long-term bonds remain constant over time. We denote by $\lambda_S \in (0, 1)$ and $\lambda_L = 1 - \lambda_S$ the relative face values of the two bonds. Moreover, we denote by $\Delta q \equiv q_S - q_L$ the price differential (per unit of face value) between the two bonds.

Restructuring. Upon default, the government offers recovery rates $\mathbf{w} = (w_S, w_L)$ and selects one voting procedure to implement the restructuring. We denote by $v_i(r)$ the reservation value

⁴²A proper analysis of the optimal design of aggregated CACs, allowing for ex-ante efficiency and incentive considerations, goes beyond the scope of this paper. Section 6 discusses the pricing impact of CACs.

⁴³The bonds' Macaulay durations evaluated at rate r are given by $D_i = 1/(r + \delta_i)$.

of an investor with discount rate r holding bond i , the exact specification of which depends on the details of the microfoundation. In what follows, we use the functional form

$$v_i(r) = \frac{c_i}{r + \delta_i + \chi}, \quad \chi \geq 0.$$

Intuitively, the parameter χ captures the extent to which the investor's reservation value is discounted relative to her subjective valuation $c_i/(r + \delta_i)$ of the bond's promised cashflows. An investor holding bond i accepts the restructuring offer w_i if and only if $r \geq c_i/w_i - (\delta_i + \chi)$.

Sorting Stage. Prior to the occurrence of a restructuring with anticipated recovery rate w_i , the valuation of bond i (per unit of face value) by investor r is given by

$$Q_i(r, w_i) = \frac{c_i + \eta w_i}{r + \delta_i + \eta}.$$

Given anticipated recovery rates $\mathbf{w}$, investors choose which bond to hold based on their subjective valuations and the price differential Δq . The set of investors sorting into bond S is

$$\mathcal{R}_S(\Delta q, \mathbf{w}) = \{r \in \mathcal{R} : Q_S(r, w_S) - Q_L(r, w_L) \geq \Delta q\},$$

while investors with discount rates in $\mathcal{R}_L(\Delta q, \mathbf{w}) = \mathcal{R} \setminus \mathcal{R}_S(\Delta q, \mathbf{w})$ hold bond L . We proceed under the assumption that investors take a unit position in either bond.⁴⁴ Given a sorting partition $(\mathcal{R}_S, \mathcal{R}_L)$, the mass of investors who hold bond i is

$$\mu_i = \int_{r \in \mathcal{R}_i} dG(r),$$

and the CDF of reservation values for bond i is

$$F_i(w) = \frac{1}{\mu_i} \int_{r \in \mathcal{R}_i} \mathbf{1}_{\{v_i(r) \leq w\}} dG(r).$$

Equilibrium. An equilibrium consists of a price differential Δq^* and a partition $(\mathcal{R}_S, \mathcal{R}_L)$, along with a modification method (two-limb or single-limb) and a pair of recovery rates $\mathbf{w}^*$, such that (i) the government chooses the modification method and restructuring offers optimally given the implied distributions F_S and F_L ; (ii) investors optimally choose which bond to hold; and (iii) the market clears for each bond, $\mu_i = \lambda_i$ for $i \in \{S, L\}$.

In what follows, we restrict our attention to parametric configurations that give rise to a threshold-type equilibrium in which long-term bonds are held by the more patient investors, with discount rates below some threshold $\hat{r}$, i.e., $\mathcal{R}_L = [r_{\min}, \hat{r}]$ and $\mathcal{R}_S = [\hat{r}, r_{\max}]$. Market

⁴⁴A more general setting where heterogeneous investors have downward-sloping bond demand functions, as in preferred-habitat models of the term structure (e.g., [Vayanos and Vila \(2021\)](#)), would deliver bond-specific reservation value distributions that are qualitatively similar to ours.

clearing requires that the threshold $\hat{r}$ be such that $G(\hat{r}) = \lambda_L$. In turn, the price differential is pinned down as $\Delta q^* = Q_S(\hat{r}, w_S^*) - Q_L(\hat{r}, w_L^*)$, where w_i^* denotes the optimal offer to bond i .⁴⁵ At the restructuring stage, the reservation value distributions for bonds S and L have supports $[v_S(r_{\max}), v_S(\hat{r})]$ and $[v_L(\hat{r}), v_L(r_{\min})]$, respectively.⁴⁶

6.2 Application to the Argentine Restructuring

In this section, we take the augmented, dynamic model to the data, calibrating it to the Argentine restructuring of Macri bonds discussed in Section 4.2. In doing so, we assume that the distribution G is uniform on $[r_{\min}, r_{\max}]$ and that coupon rates are such that $c_i = \mathbb{E}(r) + \delta_i$, implying that, absent default risk, the average investor would value the bonds at par. We set the voting thresholds at $\tau_2^s = 1/2$, $\tau_2^a = 2/3$, and $\tau_1 = 3/4$, in line with the standard version of ICMA CACs that applied to Macri bonds. In the data, we group the dollar-denominated Macri bonds into two subsets, based on the differentiated offers that they received in the restructuring: the bonds of maturities 2020-2036 map into the short-term bond (S), while bonds of longer maturities 2046-2117 map into the long-term bond (L).

Table 1: Calibration Targets.

Variable description	Model counterpart	Data Value
Relative size of short-term bond (%)	λ_S	76.0
Short-term bond duration	D_S	3.9
Long-term bond duration	D_L	9.4
Market recovery rate for short-term bond (%)	w_S	45.0
Differential recovery rate (%)	$w_S/w_L - 1$	7.0
Pre-restructuring price of short-term bond (%)	q_S	45.8
Pre-restructuring bond price differential (%)	$q_S/q_L - 1$	12.5

Table 1 displays the calibration targets: the relative size of the shorter maturity bonds, the duration of short- and long-maturity bonds, their recovery rates, and pre-restructuring prices as of March 2020.⁴⁷ For the parameter values displayed in Table 2, the model can fit these different dimensions of the data perfectly. The relative face value of short-term bonds in the data directly pins down $\lambda_S = 0.76$. Other aspects of the data imply decay rates

⁴⁵Any prices (q_S, q_L) such that $q_S - q_L = \Delta q^*$ and $q_S \leq Q_S(r_{\max}, w_S^*)$ could be observed in equilibrium. The latter inequality ensures that all investors in $\mathcal{R}_S$ are happy to buy bond S and also guarantees that $q_L = q_S - \Delta q^* \leq Q_L(\hat{r}, w_L^*)$, thus ensuring that all investors in $\mathcal{R}_L$ are happy to buy bond L .

⁴⁶Appendix E.1 shows that, when G is uniform, the reservation value CDFs and their supports (as well as the optimal restructuring offers under each aggregation method) can be written in closed form.

⁴⁷Appendix E.2 provides further details on the calibration procedure. Given the indeterminacy of bond price levels, we require parameter values to be such that $Q_S(r_{\max}, w_S^*)$ is greater than the target value for the pre-restructuring price of short-term bonds, and then consider the equilibrium where q_S matches its target and $q_L = q_S - \Delta q^*$. Model-implied bond durations (D_i) are evaluated at the threshold discount rate $\hat{r}$.

$(\delta_S, \delta_L) = (0.154, 0.004)$, default intensity and discount parameters $\eta = 0.177$ and $\chi = 0.38$, respectively, and a range of discount rates $\mathcal{R} = [0.038, 0.309]$.

Table 2: Calibrated Parameters.

Parameter	Description	Value
λ_S	Share of short-term bond	0.76
δ_S, δ_L	Decay rates	0.154, 0.004
η	Default intensity	0.177
χ	Reservation value discount parameter	0.38
$r_{\min}, r_{\max}$	Support for discount rate	[0.038, 0.309]

Figure 3 illustrates the model equilibrium objects graphically for these parameter values. The left panel of the figure depicts the bond valuation functions $Q_i(\cdot, w_i^*)$ and the threshold $\hat{r}$ in the sorting stage; the center panel plots the reservation value functions $v_i(\cdot)$, using shaded lines in the regions of $\mathcal{R}$ where they are not relevant; and the right panel illustrates the reservation value CDFs and optimal offers in the restructuring stage. As observed ahead of the Argentine default, the short-term bond commands a higher market price in the trading stage ($\Delta q^* > 0$). The holders of these bonds also tend to be more demanding in the restructuring, as seen from the relative position of the CDFs in the right panel. Optimal offers under two-limb and single-limb aggregation can be read off the circle and diamond markers, respectively. The calibrated parameter values are such that the auxiliary solution is feasible and two-limb voting dominates: the government makes differentiated offers across series, extending a more generous recovery value to the short-term bond ($w_S^* > w_L^*$). It is worth noting that, since the model matches market recovery rates and pre-restructuring bond prices perfectly, it also matches recovery rates expressed as a fraction of the pre-restructuring prices. Interestingly,

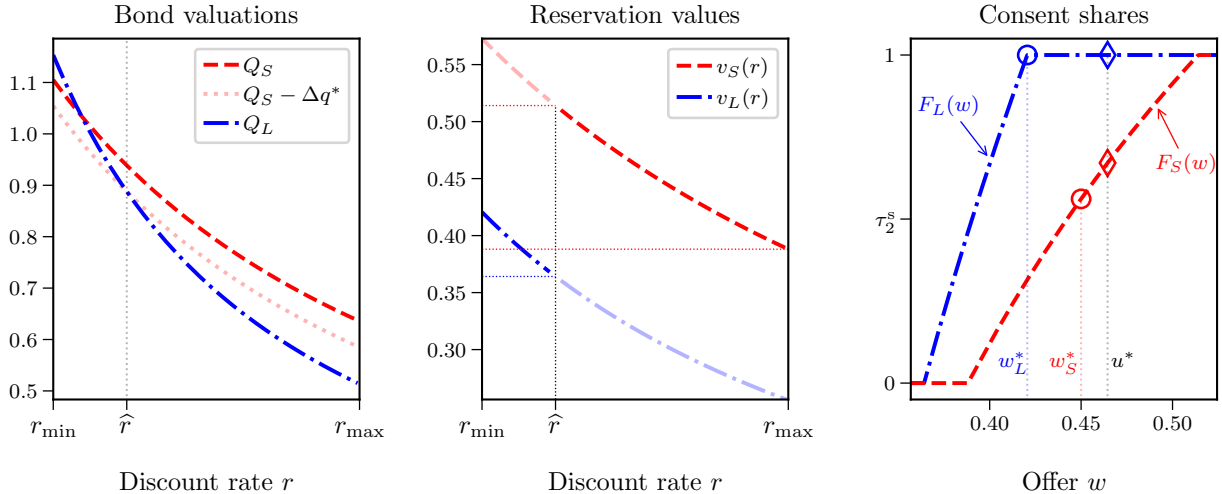


Figure 3: Equilibrium Outcomes in the Sorting and Restructuring Stages.

for Macri bonds, this measure of recovery was smaller for shorter-maturity bonds than for longer maturities, in line with the regularity documented by [Asonuma et al. \(2023\)](#).

Counterfactual Analysis: Equilibrium Impact of CAC Design. We now expand on the previous calibration to illustrate how different specifications of majority voting provisions affect equilibrium outcomes at the time of and prior to restructuring. To do so, holding parameter values fixed, we re-compute the model equilibrium assuming that, instead of enhanced ICMA CACs, the Macri bonds featured either two-limb only, single-limb only, or series-by-series CACs—with voting thresholds $\tau_2^s = 1/2$ and $\tau_2^a = 2/3$ under two-limb CACs, and a 75% supermajority under single-limb and series-by-series CACs. Table 3 displays the results.

Table 3: Counterfactual Analysis.

Outcome	Description	Two-Limb	Single-Limb	Series-by-Series
w_S	Recovery for short-term bond (%)	45.0	46.4	47.5
w_L	Recovery for long-term bond (%)	42.1	46.4	40.5
C	Total restructuring cost (%)	44.3	46.4	45.9
$F_S(w_S)$	Pre-CAC participation for short-term bond (%)	56.1	67.1	75.0
$F_L(w_L)$	Pre-CAC participation for long-term bond (%)	100.0	100.0	75.0
F_{agg}	Aggregate participation rate (%)	66.7	75.0	75.0
Δq^*	Pre-restructuring price difference (%)	5.1	2.9	6.9

At the calibrated parameter values, the outcome under two-limb CACs coincides with the one under enhanced CACs, since two-limb voting with differentiated offers allows the government to minimize the total restructuring payout. Under single-limb CACs, due to the high weight λ_S of the more demanding short-term bonds, the uniform offer ($w_S = w_L = 46.4\%$) attracts a consent share from these bonds above the per-series threshold $\tau_2^s = 1/2$, making the unique advantage of this method worthless. Interestingly, the total restructuring cost under the uniformly applicable single-limb procedure is larger than under series-by-series CACs. Under the latter, the government suffers from having to meet the 75% per-series threshold for the more demanding short-term bond, but it benefits from being able to differentiate offers across bonds. Pre-CAC participation rates vary across different CAC specifications, in line with the way the government adjusts its restructuring offers in order to meet the relevant thresholds.⁴⁸ Ahead of the restructuring, the bond price differential ($\Delta q^* = q_S - q_L$) varies across CAC specifications and is positively related to the difference in recovery values ($w_S - w_L$) across bonds. The impact of majority voting provisions on bond prices naturally increases with the likelihood of a restructuring, as illustrated by Figure E.1 in the appendix.

⁴⁸The presence of uncertainty shocks to participation rates would preserve this feature, while resulting in higher and more realistic consent shares when participation constraints hold as equalities (see Appendix B).

7 Conclusion

For over two decades, policy efforts aimed at ensuring the timely and orderly resolution of sovereign debt distress have revolved almost entirely around CACs. This paper offers the first formal analysis of the now widespread *aggregated* CACs, which allow the government to restructure multiple bonds at the same time on the basis of the aggregate consent share.

Focusing first on the best choice of aggregation procedure by a sovereign seeking to minimize the total restructuring payout under enhanced CACs, we solve for the optimal restructuring approach as a function of the heterogeneity across bonds, their relative size, and the various voting thresholds. The resolution of the “aggregation dilemma” amounts to a non-trivial comparison between two non-linear constrained minimization problems. Our analysis reveals that resorting to the novel single-limb procedure, making a uniform offer to all series, is optimal only under specific circumstances—namely, when one bond is held by investors who are particularly reluctant to take a haircut and this bond is relatively small in the restructuring pool. Our results help rationalize the two-limb approach taken by Argentina and Ecuador in August 2020, when enhanced CACs were tested for the first time.

Our work also elucidates how aggregated CACs and the anticipation of their optimal use affect vulture entry, the sorting of investors across bonds, and relative bond valuations. In particular, we show that single-limb aggregation can serve as an *off-equilibrium* deterrent when potential holdouts are looming, even though it may not be used in equilibrium. We thus provide a theoretical justification for the design of enhanced CACs, which take full advantage of the complementarity between the two aggregation procedures.

Our analysis could be extended to $N > 2$ bonds. We conjecture that the method we have developed for the two-bond case can be applied recursively to solve for the government’s optimal restructuring approach with many bonds, including the optimal use of sub-aggregation.

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A Technical Appendix for Sections 1-3

This appendix contains the proofs and derivations of all the results stated in the main text.

A.1 Proof of Proposition 1

Define the set of offers that satisfy the aggregate constraint

$$\mathcal{A}(\tau) = \left\{ \mathbf{w} : \sum_{i \in \mathcal{B}} \lambda_i F_i(w_i) \geq \tau \right\},$$

the set of offers that satisfy the series-by-series constraints

$$\mathcal{S}(\tau) = \left\{ \mathbf{w} : F_i(w_i) \geq \tau \text{ for all } i \in \mathcal{B} \right\},$$

and the set of offers that satisfy the uniform applicability condition

$$\mathcal{U} = \left\{ \mathbf{w} : w_i = w_j \text{ for all } i, j \in \mathcal{B} \right\}.$$

The restructuring cost associated with the optimal single-limb offer is

$$C_1 \equiv \min_{\mathbf{w} \in \mathcal{A}(\tau_1) \cap \mathcal{U}} \boldsymbol{\lambda} \cdot \mathbf{w} =: \boldsymbol{\lambda} \cdot \mathbf{w}^*, \quad (\text{A.1})$$

while the restructuring cost associated with the optimal two-limb offer is

$$C_2 \equiv \min_{\mathbf{w} \in \mathcal{A}(\tau_2^a) \cap \mathcal{S}(\tau_2^s)} \boldsymbol{\lambda} \cdot \mathbf{w} =: \boldsymbol{\lambda} \cdot \mathbf{w}^{**}, \quad (\text{A.2})$$

and the auxiliary problem can be formulated as

$$\widehat{C} \equiv \min_{\mathbf{w} \in \mathcal{A}(\tau_2^a)} \boldsymbol{\lambda} \cdot \mathbf{w} =: \boldsymbol{\lambda} \cdot \widehat{\mathbf{w}}.$$

Noting that $\mathcal{A}(\tau_1) \subseteq \mathcal{A}(\tau_2^a)$ since $\tau_2^a \leq \tau_1$, condition (i) follows from the fact that

$$\mathbf{w}^* \in \mathcal{S}(\tau_2^s) \Rightarrow C_1 = \min_{\mathbf{w} \in \mathcal{A}(\tau_1) \cap \mathcal{U} \cap \mathcal{S}(\tau_2^s)} \boldsymbol{\lambda} \cdot \mathbf{w} \geq C_2, \quad (\text{A.3})$$

while condition (ii) follows from the observation that

$$\widehat{\mathbf{w}} \in \mathcal{S}(\tau_2^s) \Rightarrow \mathbf{w}^{**} = \widehat{\mathbf{w}} \text{ and } C_2 = \widehat{C} \leq C_1. \quad (\text{A.4})$$

Furthermore, (A.3) and (A.4) involve a strict inequality $C_2 < C_1$ if $\mathbf{w}^{**} \notin \mathcal{U}$. ■

A.2 Proofs and Derivations for Section 2

Proof of Lemma 1. The properties of $u^* = u(\lambda, \tau_1)$ stated in the lemma follow immediately from the implicit definition, using the stochastic ordering assumption (7). ■

Proof of Lemma 2. Using Lemma 1, it is immediate to see that if $F_H^{-1}(\tau_2^s) \leq F_L^{-1}(\tau_1)$, then $u^* \geq F_H^{-1}(\tau_2^s)$ for any value of $\lambda \in (0, 1)$. If instead $F_L^{-1}(\tau_1) < F_H^{-1}(\tau_2^s)$, there exists a unique $\lambda_\dagger \in (0, 1)$ such that $u(\lambda_\dagger, \tau_1) = F_H^{-1}(\tau_2^s)$, and $u^* = u(\lambda, \tau_1) \geq F_H^{-1}(\tau_2^s)$ for all $\lambda \geq \lambda_\dagger$. ■

Remark A-1. When considering two-limb aggregation in the two-bond case, the set $\mathcal{W}$ of possible offers w_L to the holders of bond L such that the aggregate consent requirement (8) can be met for some value of w_H is defined as follows

$$\mathcal{W} = \left\{ \begin{array}{ll} \mathbb{R}^+, & \text{if } \lambda \geq \tau_2^a \\ \left(F_L^{-1} \left(\frac{\tau_2^a - \lambda}{1 - \lambda} \right), \infty \right), & \text{if } 1 - \tau_2^a \leq \lambda < \tau_2^a \\ \left(F_L^{-1} \left(\frac{\tau_2^a - \lambda}{1 - \lambda} \right), F_L^{-1} \left(\frac{\tau_2^a}{1 - \lambda} \right) \right], & \text{if } \lambda < 1 - \tau_2^a \end{array} \right\}. \quad (\text{A.5})$$

Given an offer $w_L \in \mathcal{W}$ to bond L , the offer w_H that the sovereign needs to make to holders of bond H in order to satisfy the aggregate condition (8) is given by

$$w_H = F_H^{-1} \left(\frac{\tau_2^a - (1 - \lambda)F_L(w_L)}{\lambda} \right) \equiv g(w_L). \quad (\text{A.6})$$

The function g is strictly decreasing in w_L . Furthermore, for any offers (w_H, w_L) satisfying the aggregate requirement (8), the following equivalence relationships must hold:

$$F_H(w_H) < F_L(w_L) \quad \Leftrightarrow \quad F_L(w_L) > \tau_2^a \quad \Leftrightarrow \quad F_H(w_H) < \tau_2^a. \quad (\text{A.7})$$

Remark A-2. One can show that the objective function in (10) is strictly convex when

$$\frac{d \log f_L(w_L)}{dw_L} < \frac{d \log f_H(g(w_L))}{dw_L}. \quad (\text{A.8})$$

In particular, (A.8) is satisfied if the two densities f_H and f_L are strictly decreasing. Assuming convexity, an interior solution to (10) is pinned down by the first-order condition (11).

Remark A-3. When the auxiliary solution satisfies (11), taking into account the fact that the function g defined by (A.6) depends on λ , one can show that

$$\frac{d\hat{w}_L}{d\lambda} = \left(\frac{f'_L(\hat{w}_L)}{f'_H(\hat{w}_H)} + \frac{1 - \lambda}{\lambda} \right)^{-1} \frac{F_L(\hat{w}_L) - \tau_2^a}{\lambda^2 f_L(\hat{w}_L)},$$

while

$$\frac{d\hat{w}_H}{d\lambda} = \left(1 + \frac{1 - \lambda}{\lambda} \frac{f'_H(\hat{w}_H)}{f'_L(\hat{w}_L)}\right)^{-1} \frac{F_L(\hat{w}_L) - \tau_2^a}{\lambda^2 f_L(\hat{w}_L)}.$$

In particular, in the special case where f_H and f_L are decreasing, one can see that whenever the auxiliary solution is such that $F_L(\hat{w}_L) > \tau_2^a$ (or equivalently, in view of (A.7), such that the consent share of the cheap bond is higher than for the expensive bond), a marginal increase in the relative size λ of bond H is accompanied by an improvement in the exchange offers made to both bonds.

Proof of Lemma 3. The optimal two-limb offer coincides with the auxiliary solution $\hat{\mathbf{w}}$ as long as the latter satisfies the two per-series constraints

$$F_L(\hat{w}_L) \geq \tau_2^s, \tag{A.9}$$

$$F_H(\hat{w}_H) \geq \tau_2^s. \tag{A.10}$$

The aggregate equality constraint (8) can be written as

$$F_L(\hat{w}_L) = \frac{\tau_2^a - \lambda F_H(\hat{w}_H)}{1 - \lambda}.$$

Since $F_H(\hat{w}_H) \leq 1$, it follows that

$$F_L(\hat{w}_L) \geq \frac{\tau_2^a - \lambda}{1 - \lambda},$$

where the right-hand side is decreasing in λ . In particular, the latter inequality implies that (A.9) is satisfied if $\lambda \leq (\tau_2^a - \tau_2^s)/(1 - \tau_2^s) \in (0, 1)$. Likewise, since $F_L(\hat{w}_L) \leq 1$, it follows that

$$F_H(\hat{w}_H) = \frac{\tau_2^a - (1 - \lambda)F_L(\hat{w}_L)}{\lambda} \geq \frac{\tau_2^a - (1 - \lambda)}{\lambda},$$

where the right-hand side is increasing in λ . In particular, the latter inequality implies that (A.10) is satisfied if $\lambda \geq (1 - \tau_2^a)/(1 - \tau_2^s) \in (0, 1)$.

To determine alternative conditions for the per-series constraints to be satisfied when the above sufficient conditions on λ fail, let us define $J(w) = \lambda g(w) + (1 - \lambda)w$, which is assumed to be strictly convex with an interior minimum $\hat{w}_L \in \mathcal{W}$. By construction, $J'(\hat{w}_L) = 0$, with $J'(w) < 0$ for $w < \hat{w}_L$ and $J'(w) > 0$ for $w > \hat{w}_L$. The inequality (A.9) is equivalent to $\hat{w}_L \geq F_L^{-1}(\tau_2^s) \equiv \underline{w}$, which is guaranteed to hold if $J'(\underline{w}) \leq 0$. Differentiating with respect to w , one can show that the sign of $J'(w)$ coincides with the sign of $f_H(g(w)) - f_L(w)$. Hence

the requirement $J'(\underline{w}) \leq 0$ is itself equivalent to

$$f_L(F_L^{-1}(\tau_2^s)) \geq f_H\left(F_H^{-1}\left(\frac{\tau_2^a - (1-\lambda)\tau_2^s}{\lambda}\right)\right). \quad (\text{A.11})$$

On the other hand, one can show that the inequality (A.10) is equivalent to $\widehat{w}_L \leq g^{-1}(F_H^{-1}(\tau_2^s)) = F_L^{-1}((\tau_2^a - \lambda\tau_2^s)/(1-\lambda)) \equiv \bar{w}$, which is guaranteed to hold if $J'(\bar{w}) \geq 0$. In turn, the latter requirement is equivalent to

$$f_L\left(F_L^{-1}\left(\frac{\tau_2^a - \lambda\tau_2^s}{1-\lambda}\right)\right) \leq f_H(F_H^{-1}(\tau_2^s)). \quad (\text{A.12})$$

■

Proof of Proposition 2. The first part of the proposition is an immediate corollary of Proposition 1. For the second part of the proposition, note that

$$\tau_1 > \tau_2^a \quad \Rightarrow \quad \lim_{\lambda \downarrow 0} C_1 = F_L^{-1}(\tau_1) > F_L^{-1}(\tau_2^a) = \lim_{\lambda \downarrow 0} C_2, \quad (\text{A.13})$$

hence when $\tau_1 > \tau_2^a$, two-limb voting also dominates for λ sufficiently small. ■

Proof of Proposition 3. We prove parts (i) and (ii) of the proposition separately.

Proof of Part (i). Throughout this part, we denote by $C_1(\lambda)$ and $C_2(\lambda)$ the restructuring cost under single-limb and two-limb voting, and we assume that the aggregate thresholds are identical under the two modification methods, namely, $\tau_1 = \tau_2^a = \tau^a$. It follows immediately that $\lim_{\lambda \downarrow 0} C_1(\lambda) = \lim_{\lambda \downarrow 0} C_2(\lambda) = F_L^{-1}(\tau^a)$. In the remainder of the proof, we focus on the slope of the functions C_1 and C_2 for λ close to zero. Noting that $C_1(\lambda) = u(\lambda, \tau^a)$, we apply the implicit function theorem to obtain

$$\frac{dC_1(\lambda)}{d\lambda} = \frac{\partial u(\lambda, \tau^a)}{\partial \lambda} = \frac{F_L(u^*) - F_H(u^*)}{\lambda f_H(u^*) + (1-\lambda)f_L(u^*)} > 0.$$

In particular

$$C_1'(0) = \lim_{\lambda \downarrow 0} \frac{dC_1(\lambda)}{d\lambda} = \frac{\tau^a - F_H \circ F_L^{-1}(\tau^a)}{f_L \circ F_L^{-1}(\tau^a)}. \quad (\text{A.14})$$

Next, we analyze the restructuring cost under two-limb voting, $C_2(\lambda)$. First, we establish that under the assumptions of Proposition 3 and for λ small, the individual consent requirement on bond H is binding. To see this, we start from the identity

$$f_L(F_L^{-1}(\tau^a)) - f_H(F_H^{-1}(\tau_2^s)) = \left[f_L(F_L^{-1}(\tau^a)) - f_H(F_L^{-1}(\tau^a)) \right] + \left[f_H(F_L^{-1}(\tau^a)) - f_H(F_H^{-1}(\tau_2^s)) \right].$$

By assumption, $F_L^{-1}(\tau^a) < \tilde{w}$ and $f_L(w) > f_H(w)$ for all $w < \tilde{w}$, implying that the first term is strictly positive. Noting that the second term is also strictly positive under the assumptions of the proposition, we thus conclude that

$$f_L(F_L^{-1}(\tau^a)) > f_H(F_H^{-1}(\tau_2^s)). \quad (\text{A.15})$$

In turn, the inequality (A.15) implies that (A.12) is violated for λ close to zero. Hence for λ close to zero, the two-limb offer is constrained by the individual consent requirement on bond H . Therefore, in this neighborhood,

$$C_2(\lambda) = \lambda F_H^{-1}(\tau_2^s) + (1 - \lambda) F_L^{-1}\left(\frac{\tau^a - \lambda \tau_2^s}{1 - \lambda}\right),$$

and

$$\frac{dC_2(\lambda)}{d\lambda} = F_H^{-1}(\tau_2^s) - F_L^{-1}\left(\frac{\tau^a - \lambda \tau_2^s}{1 - \lambda}\right) + \frac{\tau^a - \tau_2^s}{1 - \lambda} \left[f_L \circ F_L^{-1}\left(\frac{\tau^a - \lambda \tau_2^s}{1 - \lambda}\right) \right]^{-1}.$$

In particular,

$$C_2'(0) = \lim_{\lambda \downarrow 0} \frac{dC_2(\lambda)}{d\lambda} = F_H^{-1}(\tau_2^s) - F_L^{-1}(\tau^a) + \frac{\tau^a - \tau_2^s}{f_L \circ F_L^{-1}(\tau^a)}. \quad (\text{A.16})$$

Combining (A.14) and (A.16), we obtain that

$$C_1'(0) < C_2'(0) \quad \Leftrightarrow \quad (F_H^{-1}(\tau_2^s) - F_L^{-1}(\tau^a)) f_L(F_L^{-1}(\tau^a)) > \tau_2^s - F_H(F_L^{-1}(\tau^a)).$$

To show that this inequality holds, we note that f_H being strictly decreasing implies that F_H is strictly concave, which in turn implies that

$$(F_H^{-1}(\tau_2^s) - F_L^{-1}(\tau^a)) f_H(F_L^{-1}(\tau^a)) > F_H(F_H^{-1}(\tau_2^s)) - F_H(F_L^{-1}(\tau^a)) = \tau_2^s - F_H(F_L^{-1}(\tau^a)),$$

and the desired inequality follows from the fact that $f_L(F_L^{-1}(\tau^a)) > f_H(F_L^{-1}(\tau^a))$, as noted above. By continuity, we conclude that $C_1 < C_2$ for λ close to 0.

Proof of Part (ii). Noting that $F_H^{-1}(\tau_2^s) < F_H^{-1}(\tau_1)$, we know by assumption that the location of the crossing point $\tilde{w}$ is such that $\tilde{w} \in (F_L^{-1}(\tau_1), F_H^{-1}(\tau_1))$. In view of Lemma 1, the intermediate value theorem implies that there exists $\tilde{\lambda} \in (0, 1)$ such that $u(\tilde{\lambda}, \tau_1) = \tilde{w}$. By construction, the single-limb uniform offer u^* is equal to $\tilde{w}$ for $\lambda = \tilde{\lambda}$. Moreover, $\tilde{\lambda} F_H(\tilde{w}) + (1 - \tilde{\lambda}) F_L(\tilde{w}) = \tau_1$ is equivalent to $\tilde{w} = g(\tilde{w}; \tilde{\lambda}, \tau_1)$, where $g(w; \lambda, \tau)$ denotes the unique value of w_H such that $\lambda F_H(w_H) + (1 - \lambda) F_L(w) = \tau$. Therefore we can write

$$f_L(\tilde{w}) = f_H(\tilde{w}) = f_H(g(\tilde{w}; \tilde{\lambda}, \tau_1)). \quad (\text{A.17})$$

Setting $\lambda = \tilde{\lambda}$, we next consider the restructuring cost under two-limb voting for $\tau_2^a = \tau_1$. The assumption that the densities are decreasing guarantees convexity of the corresponding auxiliary problem, and (A.17) implies that the auxiliary solution is given by $\hat{\mathbf{w}} = (\tilde{w}, \tilde{w})$. Yet since $F_H(\tilde{w}) < \tau_2^s$ by assumption, the auxiliary solution violates the individual consent requirement on bond H , implying that $C_2 > \tilde{w}$. Therefore, for $\lambda = \tilde{\lambda}$ and $\tau_2^a = \tau_1$, we have $C_1 = \tilde{w} < C_2$ and by continuity, we conclude that the single-limb method is optimal when the parameters (λ, τ_2^a) are close to $(\tilde{\lambda}, \tau_1)$. ■

A.3 Derivations and Additional Results for Section 3

With a view towards applying some of the results derived in Section 2, we first note that under the parametric specification (12), the two densities are strictly decreasing, with

$$f_i(w) = \frac{1}{\phi_i} e^{-\frac{w}{\phi_i}}.$$

Moreover, the two densities cross at a single point $\tilde{w}$ given by

$$\tilde{w} = \frac{\gamma \log \gamma}{\gamma - 1} \phi_L. \quad (\text{A.18})$$

Holding ϕ_L fixed, $\tilde{w}$ is increasing in the heterogeneity parameter γ . Furthermore, we note that $F_i^{-1}(\tau) = -\phi_i \log(1 - \tau)$, for all $\tau \in (0, 1)$, and

$$F_L^{-1}(\tau_1) < F_H^{-1}(\tau_2^s) \quad \Leftrightarrow \quad \gamma > \frac{\log(1 - \tau_1)}{\log(1 - \tau_2^s)} \equiv \bar{\gamma}_U \quad (> 1). \quad (\text{A.19})$$

Auxiliary Solution. Under the exponential specification, condition (A.8) for strict convexity of the auxiliary problem is satisfied and the mapping g that captures the aggregate consent requirement becomes

$$g(w_L) = -\phi_H \log \left(\frac{1 - \tau_2^a - (1 - \lambda)e^{-w_L/\phi_L}}{\lambda} \right), \quad w_L \in \mathcal{W}.$$

The first-order condition (11) to the auxiliary problem gives⁴⁹

$$\hat{w}_L = \phi_L \log \left(\frac{1 + \lambda(\gamma - 1)}{1 - \tau_2^a} \right), \quad (\text{A.20})$$

⁴⁹The solution to the auxiliary problem is pinned down by this condition as long as the problem's solution is interior. In view of (A.5), one can see that the auxiliary solution is non-interior if and only if $\lambda < 1 - \tau_2^a$ and $\gamma > (1 - \lambda)/(1 - \lambda - \tau_2^a)$, in which case $\hat{w}_L = \sup \mathcal{W}$ and the consent share for bond H is zero, thus violating the individual constraint for this bond. Hence, whenever the optimal two-limb offers coincide with the auxiliary solution, the recovery values on the two bonds are given by (A.20) and (A.21).

and the induced consent share for bond L is given by (13). Since $F_L(\hat{w}_L) > \tau_2^a$, it follows from (A.7) that $F_H(\hat{w}_H) < \tau_2^a$. Hence, in the context of this parametric example, only the individual constraint on bond H may ever be binding. We compute

$$\hat{w}_H = g(\hat{w}_L) = \phi_H \log \left(\frac{1 + \lambda(\gamma - 1)}{\gamma(1 - \tau_2^a)} \right), \quad (\text{A.21})$$

and the consent share for bond H is given by (14). The auxiliary offers $\hat{w}_L$ and $\hat{w}_H$, and the corresponding consent shares, are increasing in λ —in line with the more general result stated in Remark A-3 in Appendix A.2. Moreover, one can check that $F_L(\hat{w}_L)$ is increasing in γ while $F_H(\hat{w}_H)$ is decreasing in γ —that is, the spread in consent shares under the auxiliary solution is increasing in the degree of bond heterogeneity.

Optimal Two-Limb Offer. The optimal offers under two-limb voting coincide with the auxiliary solution (A.20)–(A.21) as long as $F_H(\hat{w}_H) \geq \tau_2^s$. Using (14), one can see that the latter inequality is equivalent to

$$\frac{1 + \lambda(\gamma - 1)}{\gamma} \geq \frac{1 - \tau_2^a}{1 - \tau_2^s}. \quad (\text{A.22})$$

For given values of the voting thresholds τ_2^a and τ_2^s , one can characterize more explicitly the set of values for λ and γ such that (A.22) holds. Since $F_H(\hat{w}_H)$ is increasing in λ and decreasing in γ , the inequality is more likely to hold for high values of λ and low values of γ . Indeed, it is easy to see that (A.22) holds

- if $\gamma \leq (1 - \tau_2^s)/(1 - \tau_2^a) \equiv \bar{\gamma}_X$, for all values of λ ;
- if $\lambda \geq (1 - \tau_2^a)/(1 - \tau_2^s) \equiv \underline{\lambda} \in (0, 1)$, for all values of γ ;
- in the remainder of the parameter space for γ sufficiently small or λ sufficiently large (see Remark A-4 below for an explicit condition).

Conversely, the consent requirement on the expensive bond H is binding when there is sufficient heterogeneity across the two bonds ($\gamma > \bar{\gamma}_X$) and the relative size of bond H is small. For such parameter values, the optimal two-limb offers are

$$w_H = F_H^{-1}(\tau_2^s) = \phi_H \log \left(\frac{1}{1 - \tau_2^s} \right), \quad (\text{A.23})$$

$$w_L = F_L^{-1} \left(\frac{\tau_2^a - \lambda \tau_2^s}{1 - \lambda} \right) = \phi_L \log \left(\frac{1 - \lambda}{1 - \tau_2^a - \lambda(1 - \tau_2^s)} \right). \quad (\text{A.24})$$

Remark A-4. When $\gamma > \bar{\gamma}_X = \underline{\lambda}^{-1}$, the condition on λ such that (A.22) holds is

$$\lambda \geq \frac{\underline{\lambda}\gamma - 1}{\gamma - 1} \equiv \ell_X(\gamma) \in (0, 1), \quad (\text{A.25})$$

where ℓ_X is increasing in γ and converges to $\underline{\lambda}$ in the limit as γ goes to infinity. When $\lambda < \underline{\lambda}$, the condition on γ can be expressed as

$$\gamma \leq \frac{1 - \lambda}{\underline{\lambda} - \lambda}, \quad (\text{A.26})$$

where the right-hand side is increasing in λ , starting at $\bar{\gamma}_X$ for $\lambda = 0$ and going to infinity in the limit as $\lambda \uparrow \underline{\lambda}$.

Optimal Voting Procedure. When parameter values are such that (A.22) holds, two-limb aggregation is optimal since the series-by-series constraints have no bite. Yet another sufficient condition for two-limb optimality is that the optimal uniform offer u^* satisfies the series-by-series constraints, in which case the unique advantage of single-limb aggregation is worthless. Applying Lemma 2 (see also Proposition 2), one can see that $F_H(u^*) \geq \tau_2^s$ if $\gamma \leq \bar{\gamma}_U$, where $\bar{\gamma}_U$ is given by (A.19), or alternatively if

$$\lambda \geq \frac{1 - \tau_1 - (1 - \tau_2^s)^\gamma}{1 - \tau_2^s - (1 - \tau_2^s)^\gamma} \equiv \ell_U(\gamma) \in (0, 1) \quad \text{for } \gamma > \bar{\gamma}_U, \quad (\text{A.27})$$

where one can check that the definition of ℓ_U ensures that $u(\ell_U(\gamma), \tau_1) = F_H^{-1}(\tau_2^s)$. It is worth noting that Condition (A.27) can also be viewed as setting an upper bound on γ that is an increasing function of λ , starting at $\bar{\gamma}_U$ for $\lambda = 0$ and going to infinity as $\lambda \uparrow (1 - \tau_1)/(1 - \tau_2^s)$. One can see that when $\tau_1 = \tau_2^a$, the functions ℓ_U^{-1} and ℓ_X^{-1} both have an asymptote at $\lambda = \underline{\lambda}$, whereas if $\tau_1 > \tau_2^a$ the asymptote for ℓ_U^{-1} is at a value of λ strictly below $\underline{\lambda}$.

In order to exploit Proposition 3 and establish instead a sufficient condition for single-limb optimality, we first need to find restrictions that ensure that $\tilde{w} \in (F_L^{-1}(\tau_1), F_H^{-1}(\tau_2^s))$. It is immediate to see from (A.18) that this amounts to finding $\gamma > \bar{\gamma}_U$ such that

$$-\log(1 - \tau_1) < \frac{\gamma \log \gamma}{\gamma - 1} < -\gamma \log(1 - \tau_2^s), \quad (\text{A.28})$$

and one can show (see Remark A-5 below) that these inequalities are satisfied for high values of γ . For such values of γ , we know from Proposition 3 that single-limb voting is optimal for λ sufficiently close to zero when $\tau_1 = \tau_2^a$, and in the neighborhood of

$$\tilde{\lambda} = \frac{1 - \tau_1 - \gamma^{-\frac{\gamma}{\gamma-1}}}{(\gamma - 1)\gamma^{-\frac{\gamma}{\gamma-1}}}, \quad (\text{A.29})$$

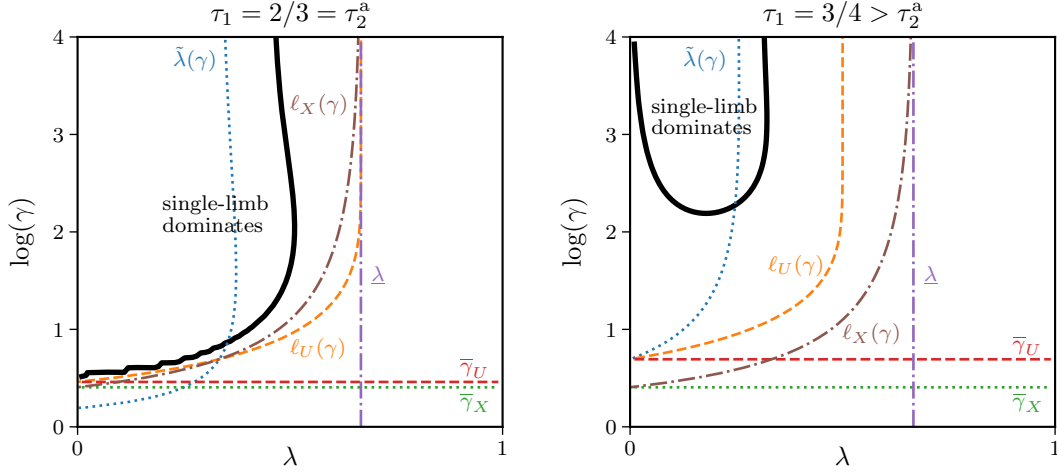


Figure A.1: Optimal Voting Rule as a function of Relative Size (λ) and Bond Heterogeneity (γ).

Notes: in each panel, the solid black line depicts the frontier delineating regions of the parameter space where single-limb or two-limb voting dominates, based on the *necessary and sufficient* condition (A.30). The other lines are related to *sufficient* conditions for one or the other method to dominate. Conditions $\gamma \leq \bar{\gamma}_X$, $\lambda \geq \Delta$, or $\lambda \geq \ell_X(\gamma)$ guarantee that $F_H(\hat{w}_H) \geq \tau_2^s$, while conditions $\gamma \leq \bar{\gamma}_U$ or $\lambda \geq \ell_U(\gamma)$ guarantee that $F_H(u^*) \geq \tau_2^s$. In line with Part (ii) of Proposition 3, single-limb dominates for γ sufficiently large and λ close to $\tilde{\lambda}(\gamma)$. The notations involved in all these sufficient conditions are explicitly defined in the text of the appendix.

when $\tau_1 - \tau_2^a$ is small, where one can check that the point $\tilde{\lambda} > 0$ is such that $u(\tilde{\lambda}, \tau_1) = \tilde{w}$.

Remark A-5. To see that the inequalities in (A.28) are satisfied for high values of γ , first note that $-\log(1 - \tau) > 0$ is increasing in τ , with $-\log(1 - \tau) > 1$ for $\tau > 1 - e^{-1} \approx 0.63$. Since $\gamma \log \gamma / (\gamma - 1)$ is strictly increasing in γ , with limit 1 as $\gamma \downarrow 1$ and going to infinity as $\gamma \rightarrow \infty$, the first inequality is satisfied for γ sufficiently large. Likewise, since $\log \gamma / (\gamma - 1)$ is strictly decreasing in γ , with limit 1 as $\gamma \downarrow 1$ and going to zero as $\gamma \rightarrow \infty$, the second inequality is also satisfied for γ sufficiently large.

Necessary and Sufficient Condition. When none of the sufficient conditions holds, the participation constraint on bond H must be binding under two-limb voting, in which case the optimal two-limb offers are given by (A.23)-(A.24). The restructuring cost is then given by

$$C_2 = \lambda \phi_H \log \left(\frac{1}{\zeta_2^s} \right) + (1 - \lambda) \phi_L \log \left(\frac{1 - \lambda}{\zeta_2^a - \lambda \zeta_2^s} \right),$$

where $\zeta_2^j = 1 - \tau_2^j$ for $j \in \{s, a\}$. Under single-limb voting, the optimal uniform offer u^* is implicitly defined by

$$F(u^*) \equiv \lambda F_H(u^*) + (1 - \lambda) F_L(u^*) = 1 - \lambda e^{-u^*/\phi_H} - (1 - \lambda) e^{-u^*/\phi_L} = \tau_1,$$

and the total cost for the government is $C_1 = u^* = F^{-1}(\tau_1)$. Single-limb aggregation is optimal if and only if $C_1 = F^{-1}(\tau_1) < C_2$, which is equivalent to $\tau_1 < F(C_2)$. We thus obtain

a necessary and sufficient condition for single-limb optimality in terms of parameter values:

$$\lambda (\zeta_2^s)^\lambda \left(\frac{\zeta_2^a - \lambda \zeta_2^s}{1 - \lambda} \right)^{\frac{1-\lambda}{\gamma}} + (1 - \lambda) (\zeta_2^s)^{\lambda\gamma} \left(\frac{\zeta_2^a - \lambda \zeta_2^s}{1 - \lambda} \right)^{1-\lambda} < 1 - \tau_1. \quad (\text{A.30})$$

Figure A.1 represents regions of the parameter space where each voting procedure dominates.

B Extension with Stochastic Consent Shares

This appendix extends the analysis of the two-bond case presented in Sections 2 and 3 to consider a situation where the government faces some uncertainty on the consent shares that a given restructuring offer may attract. We first consider a general formulation of the government's problem in the presence of uncertainty, and then provide a parametric example.

Assumptions. We proceed under the following assumptions. First, the share of consent among the holders of each bond is a random variable, whose distribution depends on the recovery rate offered to that bond. For now, we leave this distribution and its dependence on the government's offer unspecified, and allow for the possibility that consent shares may be correlated across bonds. Second, we assume that, if one or more series cannot be restructured through the activation of CACs, the bonds of these series are left unrestructured (i.e., the minimum participation threshold is no smaller than the CAC thresholds) and the government incurs a pecuniary cost Z per unit of face value of the unrestructured series.

Under single-limb aggregation, the presence of a unique constraint implies that either both bonds are restructured, or none of them is. Under the two-limb procedure, there is the possibility that the consent shares satisfy both the aggregate constraint and the series-by-series constraint for one bond, but not that for the other bond. In this case, we assume the presence of a redesignation clause, implying that the latter bond drops out of the restructuring pool and is left unrestructured, while the former is restructured through the triggering of CACs.

Notation. We denote by $\boldsymbol{\tau} \equiv (\tau_2^a, \tau_2^s)$ the pair of thresholds under the two-limb procedure. To formulate the problem of the sovereign in the presence of uncertainty, we denote by $p_a(\mathbf{w}, \tau)$ the probability that a restructuring offer $\mathbf{w} = (w_H, w_L)$ attracts an aggregate consent share above the generic aggregate threshold τ , with $p_a(u, \tau) \equiv p_a((u, u), \tau)$; we denote by $p_{HL}(\mathbf{w}, \boldsymbol{\tau})$ the probability that the offer $\mathbf{w}$ attracts consent shares that satisfy *all* constraints under the two-limb voting rule, i.e. both the series-by-series constraints and the aggregate constraint; and we denote by $p_i(\mathbf{w}, \boldsymbol{\tau})$ the probability that *only* bond i is restructured via redesignation under two-limb, i.e. the aggregate constraint and the series-by-series constraint for bond i are satisfied, but the series-by-series constraint for bond $j \neq i$ is not satisfied. The probability

that both bonds are left unrestructured under the two-limb procedure is

$$p_0(\mathbf{w}, \boldsymbol{\tau}) \equiv 1 - p_{HL}(\mathbf{w}, \boldsymbol{\tau}) - \sum_i p_i(\mathbf{w}, \boldsymbol{\tau}).$$

Government's Problem. Let $\boldsymbol{\lambda} = (\lambda_H, \lambda_L)$ denote the relative sizes of the two bonds. The cost-minimization problem of the government under two-limb aggregation is given by

$$\mathbb{E}[C_2] = \min_{\mathbf{w}} \left\{ p_{HL}(\mathbf{w}, \boldsymbol{\tau}) (\boldsymbol{\lambda} \cdot \mathbf{w}) + \sum_i p_i(\mathbf{w}, \boldsymbol{\tau}) (\lambda_i w_i + (1 - \lambda_i)Z) + p_0(\mathbf{w}, \boldsymbol{\tau})Z \right\}.$$

Instead, under the single-limb procedure, the government's problem is given by

$$\mathbb{E}[C_1] = \min_u \left\{ p_a(u, \tau_1)u + (1 - p_a(u, \tau_1))Z \right\}.$$

Special Case. In one particular specification of the model with stochastic consent shares, an offer w_i to bond i attracts a consent share $F_i(w_i) - \nu_i$, where the noise terms (ν_H, ν_L) are distributed according to the multivariate standard normal distribution

$$\begin{bmatrix} \nu_H \\ \nu_L \end{bmatrix} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}), \quad \text{where} \quad \boldsymbol{\mu} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \boldsymbol{\Sigma} = \begin{bmatrix} \sigma^2 & \rho \sigma^2 \\ \rho \sigma^2 & \sigma^2 \end{bmatrix}.$$

It immediately follows that the expected consent shares are given by $\mathbb{E}[F_i(w_i) - \nu_i] = F_i(w_i)$. Under this specification, the aggregate consent share is $\sum_i \lambda_i (F_i(w_i) - \nu_i)$, implying that

$$p_a(\mathbf{w}, \boldsymbol{\tau}) = \Phi \left(\frac{\sum_i \lambda_i F_i(w_i) - \tau}{\sigma \sqrt{\lambda_H^2 + \lambda_L^2 + 2\rho \lambda_H \lambda_L}} \right). \quad (\text{B.1})$$

Under two limb, all consent requirements are satisfied when the following conditions hold:

$$\begin{aligned} F_i(w_i) - \tau_2^s &\geq \nu_i, \\ F_j(w_j) - \tau_2^s &\geq \nu_j, \\ F_i(w_i) + \frac{\lambda_j F_j(w_j) - \tau_2^a - \lambda_j \nu_j}{\lambda_i} &\geq \nu_i, \end{aligned}$$

for $i, j \in \{H, L\}$, $j \neq i$. Instead, the restructuring only includes bond i and leaves bond j unrestructured in case the second inequality is reversed. It follows that

$$\begin{aligned} p_{HL}(\mathbf{w}, \boldsymbol{\tau}) &= \Pr \left(\nu_i \leq \min \left\{ F_i(w_i) - \tau_2^s, F_i(w_i) + \frac{\lambda_j F_j(w_j) - \tau_2^a - \lambda_j \nu_j}{\lambda_i} \right\} \wedge \nu_j \leq F_j(w_j) - \tau_2^s \right), \\ p_i(\mathbf{w}, \boldsymbol{\tau}) &= \Pr \left(\nu_i \leq \min \left\{ F_i(w_i) - \tau_2^s, F_i(w_i) + \frac{\lambda_j F_j(w_j) - \tau_2^a - \lambda_j \nu_j}{\lambda_i} \right\} \wedge \nu_j > F_j(w_j) - \tau_2^s \right). \end{aligned}$$

Remark. We use the above additively separable specification to have simple expressions, even though it implies that consent shares can in principle lie outside the unit interval. This can be readily fixed by assuming that consent shares are instead given by

$$\begin{aligned}\widehat{F}_a(\mathbf{w}) &= \min \left\{ \max \left\{ \sum_i \lambda_i (F_i(w_i) - \nu_i), 0 \right\}, 1 \right\}, \\ \widehat{F}_i(\mathbf{w}) &= \min \{ \max \{ F_i(w_i) - \nu_i, 0 \}, 1 \}.\end{aligned}$$

This formulation delivers probabilities p_a , p_i , p_{HL} that are identical to those specified above, but at the cost of more involved expressions.

Illustration. We now illustrate the results of the model assuming that the expected consent share functions F_i are as per (12), with the same parameters (ϕ_L, ϕ_H, γ) as the ones set in Section 3.2. The CAC thresholds are $\tau_1 = \tau_2^a = 2/3$ and $\tau_2^s = 1/2$, and the cost of leaving a bond unrestructured is $Z = 7$. Figures B.1-B.3 illustrate comparative statics with respect to the relative size (λ) of the expensive bond H , the uncertainty about consent shares (σ), and their correlation (ρ), respectively. Unless specified otherwise, we set $\sigma = 0.05$ and $\rho = 0$.

In each figure, the top left panel depicts restructuring offers under two-limb voting (dashed red for bond H and dash-dotted blue for bond L) and the uniform offer under single-limb voting (dotted black). The top right panel depicts, for each aggregation method, the expected consent shares for each bond, along with the expected aggregate consent share (with the solid orange line corresponding to the expected aggregate consent under two-limb voting). The bottom right panel depicts as a dotted black line the probability $p_a(u^*, \tau_1)$ that the restructuring goes through under single-limb, and as a solid orange line the probability $p_{HL}(\mathbf{w}, \boldsymbol{\tau})$ that the activation of CACs allows the restructuring of both bonds under two-limb voting, as well as the probabilities $p_i(\mathbf{w}, \boldsymbol{\tau})$ that only bond $i = H, L$ is being restructured. Finally, the bottom left panel depicts the expected restructuring costs $\mathbb{E}[C_2]$ and $\mathbb{E}[C_1]$ under the two-limb and single-limb procedures (solid orange and dotted black lines, respectively).

Figure B.1 demonstrates that the findings illustrated in Figure 1 on the optimal restructuring approach generalize to a setting where the government faces uncertainty about consent shares. Optimal offers under either aggregation method are increasing in the share of the more demanding bond. In the presence of uncertainty, offers are made to ensure that, in expectation, consent shares are comfortably above the required thresholds, yet leaving the possibility that some bond(s) may be left unrestructured. For low values of λ , the optimal uniform offer under single limb is expected to attract from bond H a consent share below the series-by-series threshold τ_2^s , whereas the optimal offer to this bond under two limb is sufficiently generous to ensure that, in all likelihood, the series-by-series requirement will be met. For low values of λ , the unique advantage of single-limb voting thus makes it optimal.

Naturally, as illustrated in the top-left panel of Figure B.2, restructuring offers—as well

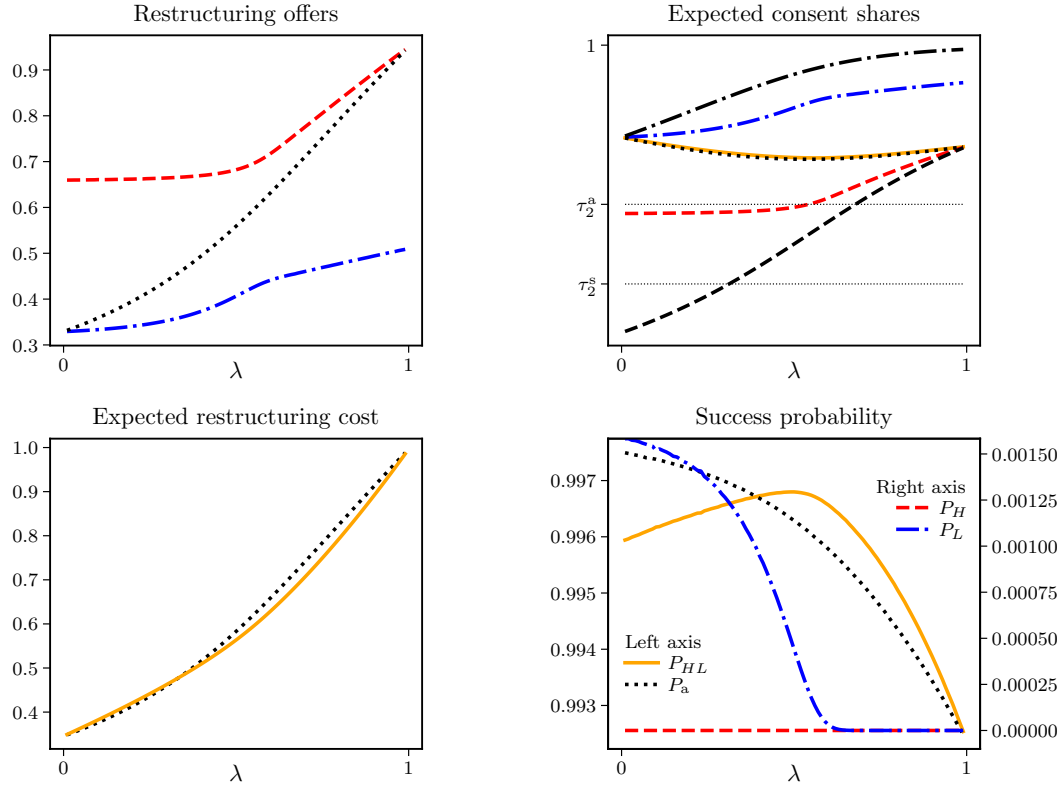


Figure B.1: Comparative Statics with Respect to λ .

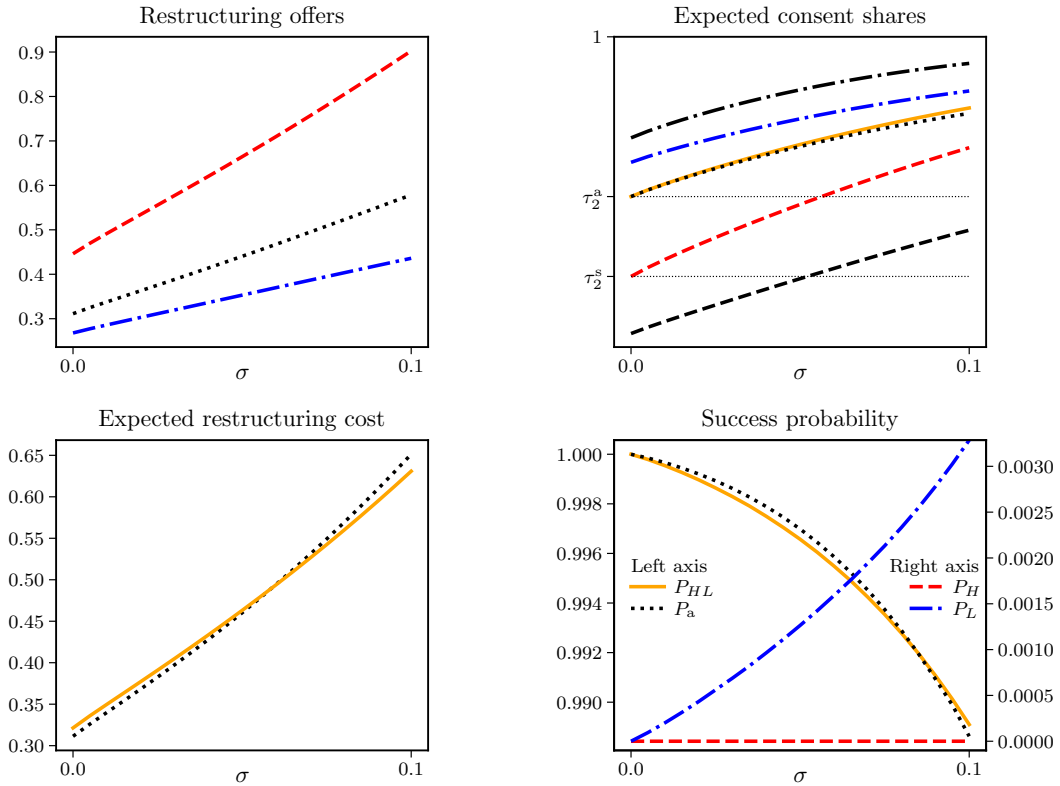


Figure B.2: Comparative Statics with Respect to σ , for $\lambda = 0.3$.

as expected consent shares—are increasing in the degree of uncertainty. As $\sigma \downarrow 0$, offers can be tailored to just meet each of the consent requirements as an equality. When instead there is some amount of uncertainty, offers are made more generous to guarantee a wider comfort margin. Despite the optimal offers being more generous, an increase in uncertainty lowers the probability of a smooth restructuring. The bottom left panel of the figure reveals that an increase in uncertainty is favorable to the use of two-limb aggregation.

Figure B.3 illustrates the impact of the correlation of shocks to consent shares on the optimal restructuring approach. One may think of this correlation as being driven (at least partly) by the extent to which some creditors are common across the bonds. As can be seen from Equation (B.1), holding everything else constant, an increase in ρ lowers the probability of reaching a given aggregate threshold. Under single limb, despite a slight upward adjustment in the uniform offer, an increase in ρ is accompanied by an approximately linear drop in the likelihood of a smooth restructuring. Under two-limb voting, the drop is less pronounced and the probability p_{HL} of a smooth restructuring even reverts back up for high values of ρ due to the adjustment in the government's offers. Overall, an increase in the correlation of shocks to consent shares contributes to making the two-limb procedure more appealing.

Finally, Figure B.4 generalizes and extends the results from Sections 2 and 3 on the determinants of the optimal aggregation method. Consistent with Figure 2 (left panel, for $\tau_1 = \tau_2^a = 2/3$), even in the presence of shocks, the single-limb procedure is found to be

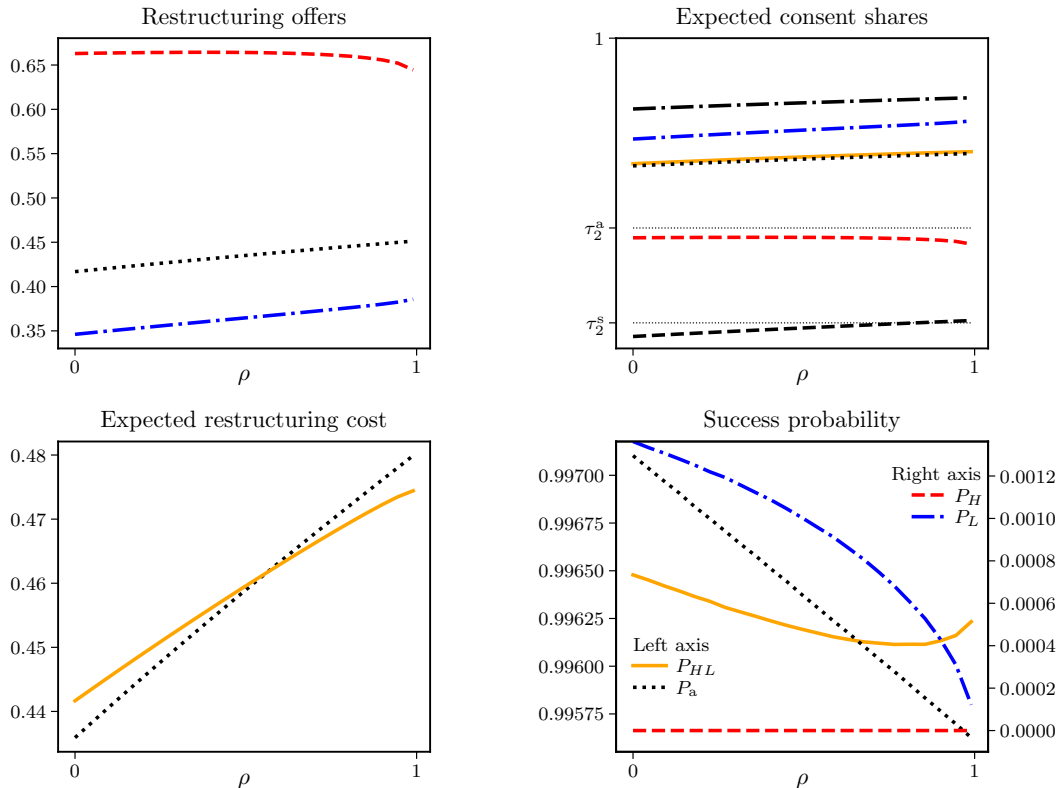


Figure B.3: Comparative Statics with Respect to ρ , for $\lambda = 0.25$.

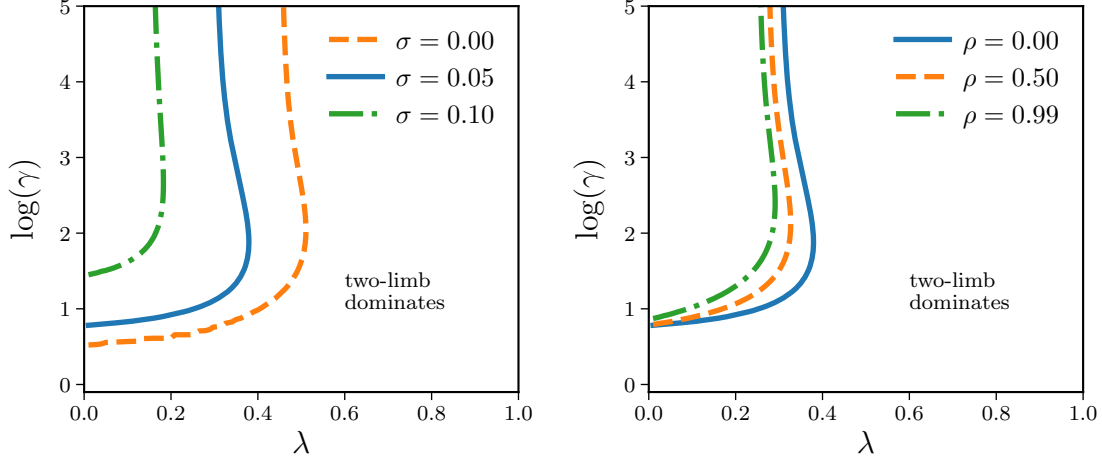


Figure B.4: Optimal Aggregation Method as a Function of Relative Size (λ) and Bond Heterogeneity (γ), for Different Levels of Volatility (σ) and Correlation (ρ) of Shocks to Consent Shares.

optimal only when there is substantial heterogeneity across bonds and the more demanding bond is relatively small. Moreover, consistent with the insights from Figures B.2 and B.3, an increase in uncertainty and in the correlation of shocks on consent shares are both favorable to the use of two-limb aggregation.

C Data

This appendix provides further details on the restructurings implemented by Greece in 2012 and by Argentina and Ecuador in 2020, which we discuss in Section 4. For each exchange, we describe (i) the set of Eligible Bonds to be restructured and their characteristics; (ii) the terms of the exchange offer, i.e., the characteristics, amount, and value of New Bonds to be received by each Eligible Bond holder in the exchange; and (iii) the results of the exchange in terms of consent shares and recovery rates.

C.1 Greece

Zettelmeyer et al. (2013) (hereafter ZTG) offer a detailed analysis of the 2012 Greek restructuring. Rather than replicating their tables, we refer here to the parts of their paper that specifically describe each aspect of the exchange.

Eligible Bonds. The amounts and characteristics of the Eligible Bonds, grouped by governing law, are described in Appendix 2 of ZTG, specifically in Tables A3 and A4. The first part of Section 3 in ZTG, including Table 2, provides information on the composition and estimated bond holdings of the creditors that comprised the Creditor Committee. Our statement on the relative price of short- vs long-maturity eligible bonds prior to restructuring refers to

the prices of bonds with a residual maturity of three months and 25 years, respectively, as observed one week before the 21-Feb-2012 press conference when the offer was announced.

Exchange Offer. Appendix 1 of ZTG describes the bundle of New Bonds in the offer.

Exchange Results. Section 3.1 in the main text of ZTG gives an overview of the outcome of the exchange offer, including details on the exit yield curve of the New Bonds. Table A4 in their Appendix 2 displays the results of the exchange in terms of consent shares (see “holdouts”) and recovery (see “haircut”), for each Eligible Bond. Only data on ex-post consent shares are available, i.e., *after* CACs were applied. Figure 5 in the main text and the associated discussion highlight the large heterogeneity in haircuts across Eligible Bonds.

C.2 Argentina

This section provides details on the 2020 Argentine restructuring discussed in Section 4.2. Given the substantial difference between Kirchner and Macri bonds, the restructuring details are reported separately for these two sets of bonds. The 2005 and 2016 indentures lay out the terms governing Kirchner and Macri bonds, respectively.

Data Sources. We use SEC’s EDGAR and the Luxembourg Stock Exchange for official documentation on exchange offers and bond prospectuses. We use LSEG Refinitiv Workspace for the historical prices and yields of the Eligible Bonds, and for the exit prices and yields of the New Bonds. We refer to exit prices and yields as those quoted for the New Bonds on 15-Sep-2020, two weeks after the completion of the exchange. To compute Eligible Bond durations, we use a reference exit yield as discount rate. Specifically, we construct a reference exit yield for each combination of Eligible Bond indenture and currency denomination, which we compute as the principal-weighted average exit yield across New Bonds of the corresponding indenture and currency. Further information on specific details of the data can be found in the companion spreadsheet.

C.2.1 Eligible Bonds

Table C.1 reports selected characteristics of the bonds issued by the Kirchner administrations as part of the 2005-2010 exchange. Par bonds had a step-up coupon structure that made their cashflow profile back-loaded. Discount bonds, on the other hand, had a fixed coupon rate, but a time-varying fraction of such coupon was capitalized, in the sense that it was not paid out and went instead to increase the face value of the bond. As a result, at the time of the 2020 exchange, the effective face value of USD and EUR Discount bonds was 140.20 and 137.96, respectively. Each Kirchner bond is split into three separate series (I, II, and III). While series I was issued as part of the 2005 debt exchange, the other two series were issued at two stages of the second debt exchange in 2010: series II was issued in the April 2010 exchange, in which

Table C.1: Eligible Series, Kirchner Bonds

Bond Name	Issue Date	Coupon Rate ^a	Maturity Date ^b	Macaulay Duration ^c	Principal Amount Outstanding (millions)	
					Owned by Argentina	Total
USD Discount 2033 I	01-Oct-2005	5.906%	31-Dec-2033	5.81	404	3,858
USD Discount 2033 II and III	28-Apr-2010	5.906%	31-Dec-2033	5.81	69	1,243
USD Par 2038 I	01-Oct-2005	step-up	31-Dec-2038	9.17	358	4,939
USD Par 2038 II and III	28-Apr-2010	step-up	31-Dec-2038	9.17	-	95
Euro Discount 2033 I	01-Oct-2005	5.668%	31-Dec-2033	5.66	7	3,108
Euro Discount 2033 II and III	21-Jun-2010	5.668%	31-Dec-2033	5.66	-	2,661
Euro Par 2038 I	01-Oct-2005	step-up	31-Dec-2038	8.90	-	5,035
Euro Par 2038 II	28-Apr-2010	step-up	31-Dec-2038	8.90	-	1,438

Notes:

^a The full coupon schedules and cashflow profiles, including principal amortization, are reported in the companion spreadsheet.

^b To compute bond durations, we use a bond-specific discount rate, as described in the preamble of Section C.2.

Brady Bond holders could not participate, while series III was issued in the December 2010 exchange directed to Brady Bond holders. Although the latter two series were given different ISINs for U.S. tax purposes, they were grouped together for all voting purposes in the 2020 restructuring and we therefore lump them together in all tables related to Kirchner bonds.

Table C.2: Eligible Series, Macri Bonds

Eligible Bond Name	Issue Date	Coupon Rate	Maturity Date	Macaulay Duration ^a	Principal Amount Outstanding (millions)	
					Owned by Argentina	Total
USD 2021	22-Apr-2016	6.875%	22-Apr-2021	0.60	16	4,484
USD 2022	26-Jan-2017	5.625%	26-Jan-2022	1.33	-	3,250
USD 2023	11-Jan-2018	4.625%	11-Jan-2023	2.22	-	1,750
USD 2026	22-Apr-2016	7.500%	22-Apr-2026	4.58	45	6,455
USD 2027	26-Jan-2017	6.875%	26-Jan-2027	5.11	-	3,750
USD 5.875% 2028	11-Jan-2018	5.875%	11-Jan-2028	5.80	-	4,250
USD 6.625% 2028	06-Jul-2016	6.625%	06-Jul-2028	5.95	35	965
USD 2036	06-Jul-2016	7.125%	06-Jul-2036	8.49	23	1,727
USD 2046	22-Apr-2016	7.625%	22-Apr-2046	9.31	132	2,618
USD 2048	11-Jan-2018	6.875%	11-Jan-2048	9.51	-	3,000
USD 2117	28-Jun-2017	7.125%	28-Jun-2117	9.42	61	2,689
Euro 2022	12-Oct-2016	3.875%	15-Jan-2022	1.32	-	1,250
Euro 2023	09-Nov-2017	3.375%	15-Jan-2023	2.27	-	1,000
Euro 2027	12-Oct-2016	5.000%	15-Jan-2027	5.34	-	1,250
Euro 2028	09-Nov-2017	5.250%	15-Jan-2028	5.91	-	1,000
Euro 2047	09-Nov-2017	6.250%	09-Nov-2047	8.48	-	750
CHF 2020	12-Apr-2017	3.375%	12-Oct-2020	0.07	-	400

Notes:

^a To compute bond durations, we use a bond-specific discount rate, as described in the preamble of Section C.2.

Table C.2 displays features of the bonds issued by the Macri administration under the 2016 indenture. This set of bonds had a more standard cashflow structure, with constant coupon rates and a bullet principal repayment at maturity. Since each bond was issued and “tapped” through conventional bond syndication, each maturity corresponds to a single series, unlike for the Kirchner bonds.

C.2.2 Exchange Offer

New Bonds. Table C.3 lists the New Bonds issued during the exchange and describes their characteristics. Interestingly, some of these bonds are governed by the 2005 indenture.

Table C.3: New Bonds Characteristics

Bond Description	Indenture	Coupon Rate ^a	Principal Amortization Payments ^b	Maturity Date	Macaulay Duration ^c	Amount Issued in the Exchange (millions)	Exit Price
USD 2038	2005	step-up	22	09-Jan-38	6.20	11,405	49.95
USD 2041	2005	step-up	28	09-Jul-41	8.79	10,482	44.45
EUR 2038	2005	step-up	22	09-Jan-38	6.02	809	56.86
EUR 2041	2005	step-up	28	09-Jul-41	8.61	1,573	52.41
USD 2030	2016	step-up	13	09-Jul-30	4.50	16,091	48.90
USD 2035	2016	step-up	10	09-Jul-35	6.82	20,502	43.22
USD 2046	2016	step-up	44	09-Jul-46	11.68	2,092	46.30
EUR 2030	2016	0.125%	13	09-Jul-30	3.08	1,166	57.42
EUR 2035	2016	step-up	10	09-Jul-35	6.85	299	52.50
EUR 2046	2016	step-up	44	09-Jul-46	10.39	248	50.18
USD 2029	2016	1%	10	09-Jul-29	3.12	2,635	52.47
EUR 2029	2016	0.5%	10	09-Jul-29	3.11	90	59.17

Notes:

^a The full coupon schedules and cashflow profiles are reported in the Online Appendix.

^b Number of equal semi-annual principal amortization payments before (and including) the maturity date.

^c To compute bond durations, we use a bond-specific discount rate, as described in the preamble to Section C.2.

Tables C.4 and C.5 describe the menu of exchange options offered to each eligible bond, showing the face value conversion rates—i.e., the amount that the holder of 100 currency units of a given Eligible Bond would receive if they exchanged it for a given New Bond. The reported conversion rates were adjusted, so that (i) they are unit-free and comparable among each other, which was done by converting the New Bond face value to the currency of the Eligible Bond using the exchange rate at the time of the offer; (ii) they take into account the higher effective face value of the Discount bonds. Eligible Bond holders who did not consent but whose bonds are being restructured nonetheless through the triggering of CACs received a fallback New Bond at the same conversion rate as consenting bondholders, indicated by a star, and a smaller amount of Arrear Bonds in compensation for arrears, indicated in parentheses.

Table C.4: Exchange Offer Menu, Kirchner Bonds

Eligible Bond Name	Principal Amount of New Bond per 100 Currency Units of Eligible Bond							
	New Bonds						Arrear Bonds	
	38 USD	41 USD	46 USD	38 EUR	41 EUR	46 EUR	29 USD	29 EUR
USD 2033 Discount I	100*	100	97				5.61 (2.58)	
USD 2033 Discount II and III	100*	100	97				5.61 (2.58)	
USD Par 2038 I		100*	97				1.6 (0.23)	
USD Par 2038 II and III		100*	97				1.6 (0.23)	
Euro 2033 Discount I	99.74			99.74*	99.74	96.75	5.29	5.29 (2.43)
Euro 2033 Discount II and III	99.74			99.74*	99.74	96.75	5.29	5.29 (2.43)
Euro Par 2038 I		100			100*	97	1.45	1.45 (0.21)
Euro Par 2038 II and III		100			100*	97	1.45	1.45 (0.21)

Note: Numbers with stars (resp. in parentheses) denote the amount of New Bonds (resp. Arrear Bonds) to be received by creditors who do not consent to the exchange, but whose bonds are restructured nonetheless through the triggering of CACs.

Table C.5: Exchange Offer Menu, Macri Bonds

Eligible Bond Name	Principal Amount of New Bond per 100 Currency Units of Eligible Bond							
	New Bonds						Arrear Bonds	
	30 USD	35 USD	46 USD	30 EUR	35 EUR	46 EUR	29 USD	29 EUR
USD 2021	97	97	97*				5.96 (3.44)	
USD 2022	97	97	97*				3.41 (1.34)	
USD 2023	97	97	97*				2.99 (1.3)	
USD 2026	97	97	97*				6.5 (3.75)	
USD 2027	97	97	97*				4.16 (1.64)	
USD 5.875% 2028	97	97	97*				3.8 (1.65)	
USD 6.625% 2028	97	97	97*				4.38 (1.95)	
USD 2036	97	97	97*				4.71 (2.1)	
USD 2046		97	97*				6.61 (3.81)	
USD 2048		97	97*				4.45 (1.93)	
USD 2117		97	97*				4.87 (2.26)	
Euro 2022	97			97	97	97*	2.47	2.47 (1.04)
Euro 2023	97			97	97	97*	2.15	2.15 (0.9)
Euro 2027		97		97	97	97*	3.18	3.18 (1.34)
Euro 2028		97		97	97	97*	3.34	3.34 (1.41)
Euro 2047			97		97	97*	5.12	5.12 (2.82)
CHF 2020	97			97	97	97*	3.02	3.02 (1.78)

Notes: Numbers with stars and in parentheses respectively denote the amount of New and Arrear Bonds to be received by creditors who do not consent to the exchange, but whose bonds are restructured nonetheless through CACs.

A few things are worth noting. First, for both sets of bonds, the holders of different Eligible Bonds were given different menus of New Bonds from which they could choose. Second, the conversion rates for Kirchner bonds were the same for all 2005-indenture New Bonds (the 38s

and 41s), but were lower for the 2016-indenture New Bonds (the 46s). Third, holders of Macri USD Eligible Bonds with maturities between 2026 and 2036 who consented to the exchange were not guaranteed their choice, depending on the choices of the holders of lower maturities. We discuss this further below (see Table C.7 and the associated text).

C.2.3 Exchange Results

Participation. Tables C.6 and C.7 show the exchange results in terms of participation for the Kirchner and Macri bonds, respectively. The left half of each table shows for each Eligible Bond series the share of creditors (i.e., the fraction of total face value) who consented to the exchange. The right half of each table shows what fraction of *consenting* holders of a given Eligible Bond tendered exchange orders for a given New Bond.

Table C.6: Exchange Results, Kirchner Bonds

Eligible Bond Name	Amount Outstanding (millions)	Consent Share (%)	% Share of Tender Orders for given New Bond					
			38 USD	41 USD	46 USD	38 EUR	41 EUR	46 EUR
USD 2033 Discount I	3,858	96.8	99.3	0.6	0.1			
USD 2033 Discount II and III	1,235	95.5	99.3	0.6	0.1			
USD Par 2038 I	4,939	94.1		99.1	0.9			
USD Par 2038 II and III	95	64.4		91.5	8.5			
Euro 2033 Discount I	3,108	89.9	93.9			5.8	0.2	0.1
Euro 2033 Discount II and III	2,661	91.3	95.7			4.3	0.1	0.0
Euro Par 2038 I	5,035	77.4		91.2			8.8	0.1
Euro Par 2038 II and III	1,438	60.5		89.9			10.0	0.1

Notes:

^a All consent shares are expressed in face value terms.

^b USD Pars II and III and EUR Pars II and III were redesignated and not restructured. Their holders were later paid in full, as announced in a press statement on 28/09/2020 by the Argentinian Ministerio de Economia.

For the Kirchner bonds, investors' choices were quite homogeneous, with at least 90% of the holders of each Eligible Bond choosing the same New Bond, and few opting for the 2016-indenture bonds. For the Macri bonds, the exchange featured "Acceptance Priority Procedures", whereby the issued quantities of New 2030 and 2035 USD Bonds was capped, and if the orders for such bonds were to exceed the cap, the holders of USD Eligible Bonds with maturities between 2021 and 2023 (both included) would be given priority, while the holders of USD Eligible Bonds with maturities between 2026 and 2036 (both included) would be assigned the requested New Bonds pro-rata, and the New Bond for which the cap was not reached for the remaining part.⁵⁰ This implies that, while the majority of those investors

⁵⁰The issuance caps for the New USD 30s and 35s were such that the issuance of only one of the two bonds might reach the cap.

Table C.7: Exchange Results, Macri Bonds

Eligible Bond Name	Amount Outstanding (millions)	Consent Share (%)	% Share of Consenting exchange orders for given New Bond ^a					
			30 USD	35 USD	46 USD	30 EUR	35 EUR	46 EUR
USD 2021	4,484	98.3	98.1	1.8	0.1			
USD 2022	3,250	86.5	96.7	3.1	0.2			
USD 2023	1,750	98.6	99.0	0.8	0.2			
USD 2026	6,455	98.5	92.4 (32.6)	7.5 (67.3)	0.1			
USD 2027	3,750	97.9	92.5 (32.6)	7.2 (67.0)	0.4			
USD 5.875% 2028	4,250	98.3	93.9 (33.2)	5.4 (66.2)	0.7			
USD 6.625% 2028	965	98.1	88.5 (31.2)	10.7 (67.9)	0.8			
USD 2036	1,727	98.8	79.3 (28.0)	19.6 (70.9)	1.0			
USD 2046	2,618	97.1		96.4	3.6			
USD 2048	3,000	97.8		97.4	2.6			
USD 2117	2,689	96.1		92.1	7.9			
Euro 2022	1,250	97.0	74.7			23.9	1.3	0.1
Euro 2023	1,000	99.6	85.5			13.5	0.6	0.5
Euro 2027	1,250	95.4		66.3		29.8	3.6	0.3
Euro 2028	1,000	98.7		62.2		35.3	2.3	0.2
Euro 2047	750	91.4			67.7		30.2	2.1
CHF 2020	400	87.6	72.6			22.7	4.5	0.2

Notes:

^a Where present, numbers in parentheses report the share of orders filled after application of the Acceptance Priority Procedures.

submitted orders for the New USD 30s, they received a bundle composed of the New USD 35s for two-thirds, and of the New USD 30s for only one-third.

Recovery Rates. Tables C.8 and C.9 display the average recovery rates for each Eligible Bond. These are computed as the average of the recovery rates obtained by exchanging the Eligible Bond for a given New Bond, weighted by the share of Eligible Bond holders who submitted orders for such New Bond, as displayed in Tables C.6 and C.7 (the companion spreadsheet shows recovery rates for each Eligible-New Bond pair allowed by the terms of the exchange). “Market” recovery rates are computed as the product of the conversion rate from Tables C.4 and C.5 and the New Bond’s exit price from Table C.3. “SZ” recovery rates follow

Table C.8: Recovery Rates, Kirchner Bonds

Eligible Bond Name	Market Recovery Rate	NPV Old Bond	SZ Recovery Rate
USD 2033 Discount I	49.9	85.5	58.4
USD 2033 Discount II and III	49.9	85.5	58.4
USD Par 2038 I	44.4	51.9	85.7
USD Par 2038 II and III	44.5	51.9	85.7
Euro 2033 Discount I	50.3	74.8	67.2
Euro 2033 Discount II and III	50.2	74.8	67.1
Euro Par 2038 I	45.2	41.6	108.7
Euro Par 2038 II and III	45.3	41.6	108.9

Notes: Recovery rates do not include the amounts of Arrear Bonds delivered as compensation for accrued interest.

Table C.9: Recovery Rates, Macri Bonds

Eligible Bond Name	Market Recovery Rate	NPV Old Bond	SZ Recovery Rate
USD 2021	47.3	97.6	48.5
USD 2022	47.3	93.1	50.7
USD 2023	47.4	86.9	54.5
USD 2026	43.7	84.9	51.5
USD 2027	43.7	80.7	54.2
USD 5.875% 2028	43.8	73.9	59.3
USD 6.625% 2028	43.7	76.6	57.0
USD 2036	43.5	70.1	62.0
USD 2046	42.0	70.0	60.0
USD 2048	42.0	63.3	66.3
USD 2117	42.2	63.6	66.3
Euro 2022	49.5	88.3	56.1
Euro 2023	48.6	79.8	60.9
Euro 2027	46.4	63.3	73.2
Euro 2028	47.0	61.0	77.1
Euro 2047	46.8	45.7	102.4
CHF 2020	49.4	102.4	48.2

Notes: Recovery rates do not include the amounts of Arrear Bonds delivered as compensation for accrued interest.

the method of [Sturzenegger and Zettelmeyer \(2006\)](#), and are obtained as the ratio of the NPV of the New Bond to the NPV of the Eligible Bond, both calculated using the New Bond's exit yield. That is, SZ recovery rates are equal to the ratio between the Market recovery rates and the NPV of the Eligible Bond. We do not include arrears in these calculations because doing so would make recovery rates quite heterogeneous for a non-fundamental reason, which is that differences in arrears across bonds are entirely due to differences in bond coupon dates.

A few things are worth noting. First, market recovery rates show that investors' exchange choices are broadly consistent with exit prices. For the Kirchner bonds in any currency, the New 38s had a higher exit price than the New 41s, so the holders of Discount Eligible Bonds had higher market recovery rates than the holders of Pars, since the former could choose the New 38s and the latter could not. For the Macri bonds, the Acceptance Priority Procedures cost the holders of the USD 2026-2036s around 5 percentage points in recovery rates. For the holders of the USD 2046-2117, opting for the New USD 46s instead of the New USD 35s would have implied approximately a 3 p.p. higher market recovery rate. Second, while market recovery rates are decreasing in the duration of the corresponding Eligible Bond, the converse holds for the SZ method. Indeed, while Argentina offered more valuable New Bonds to the holders of Eligible Bonds with shorter durations, this is more than offset by the higher NPV of those Eligible Bonds.⁵¹

⁵¹In the case of Macri bonds, this contrast would disappear if the exit yields used for discounting were approximately 5 percentage points lower.

C.3 Ecuador

This section provides further details on the Ecuadorian restructuring discussed in Section 4.3. We refer to exit prices and yields for the New Bonds as the ones quoted on 7-Sep-2020, one week after these bonds started trading. To compute Eligible Bond durations we use a reference exit yield, which we compute as the principal-weighted average yield across New Bonds.

Eligible Bonds. Table C.10 lists the Eligible Bonds and describes their features.

Table C.10: Eligible Series, Ecuador

Eligible Bond Name	Issue Date	Coupon Rate	Maturity Date	Macaulay Duration	Principal Amount Outstanding (millions USD)	
					Owned by Ecuador	Total
USD 2022	28-Jul-2016	10.75 %	28-Mar-2022	1.47	-	2,000
USD 2023	02-Jun-2017	8.75 %	02-Jun-2023	2.48	-	1,000
USD 2024 ^a	20-Jun-2014	7.95 %	20-Jun-2024	3.33	105	2,000
USD 2025	27-Sep-2019	7.875 %	27-Mar-2025	3.89	-	600
USD 2026	13-Dec-2016	9.65 %	13-Dec-2026	4.85	12	1,750
USD 9.625% 2027	02-Jun-2017	9.625 %	02-Jun-2027	5.11	-	1,000
USD 8.875% 2027	23-Oct-2017	8.875 %	23-Oct-2027	5.40	-	2,500
USD 2028	23-Jan-2018	7.875 %	23-Jan-2028	5.65	26	3,000
USD 2029	31-Jan-2019	10.75 %	31-Jan-2029	5.83	19	2,125
USD 2030	27-Sep-2019	9.5 %	27-Mar-2030	6.48	-	1,400

Notes:

^a The USD 2024 were issued under a 2014 indenture that only featured series-by-series CACs; all other bonds were governed by a 2016 indenture featuring enhanced CACs.

Exchange Offer. Table C.11 displays the New Bonds offered in the exchange, while Table C.12 describes the terms of the exchange offer. In the latter table, each cell reports the principal amount of a given New Bond (in columns) to be received by the holder of \$100 of a given Eligible Bond (in rows). Non-consenting holders of any Eligible Bond would receive, per \$100 of Eligible Bond, \$91.13 of the New USD 2040, and no amount of the Arrear Bond.

Table C.11: New Bonds Characteristics

Bond Description	Indenture	Coupon Rate ^a	Principal Amortization Payments ^b	Maturity Date	Macaulay Duration	Amount Issued in the Exchange (million USD)	Exit Price
USD 2030	2020	step-up	10	31-Jul-30	6.64	3,701	74.13
USD 2035	2020	step-up	10	31-Jul-35	10.14	8,459	62.75
USD 2040	2020	step-up	10	31-Jul-40	12.62	3,403	57.00
USD PDI 2030	2020	0%	10	31-Jul-30	7.56	1,005	49.50

Notes:

^a The full coupon schedules and cashflow profiles are reported in the Companion Spreadsheet.

^b Number of equal semi-annual principal amortization payments before (and including) the maturity date.

Table C.12: Exchange Offer, Ecuador

Eligible Bond Name	Principal Amount of New Bond per 100 USD of Eligible Bond			
	New Bonds			Arrear Bond
	New 2030	New 2035	New 2040	New PDI 2030
USD 2022	30.0	49.53	11.6	7.89
USD 2023	22.9	49.53	18.7	5.08
USD 2024	30.0	49.53	11.6	4.27
USD 2025	18.9	49.53	22.7	5.79
USD 2026	18.9	49.53	22.7	5.35
USD 9.625% 2027	18.9	49.53	22.7	5.59
USD 8.875% 2027	18.9	49.53	22.7	5.98
USD 2028	18.9	49.53	22.7	3.61
USD 2029	18.9	49.53	22.7	4.76
USD 2030	18.9	49.53	22.7	6.99

Notes: see text for details.

Exchange Results. Table C.13 shows the exchange results in terms of participation and recovery rates. The latter are computed following the same calculation method as the one described for the case of Argentina (see Section C.2.3). Market recovery rates are quite homogeneous, and weakly decreasing in Eligible Bond maturity. SZ recovery rates, instead, show no clear pattern.

Table C.13: Exchange Participation and Recovery Rates, Ecuador

Eligible Bond Name	Principal Amount Outstanding (millions USD)	Consent Share (%)	Market Recovery Rate ^a	NPV Old Bond	SZ Recovery Rate ^a
USD 2022	2,000	98.61	59.9	101.8	58.9
USD 2023	1,000	97.76	58.7	98.2	59.8
USD 2024 ^b	2,000	95.42	59.9	95.1	63.0
USD 2025	600	97.84	58.0	94.1	61.7
USD 2026	1,750	97.99	58.0	100.6	57.7
USD 9.625% 2027	1,000	98.21	58.0	100.6	57.7
USD 8.875% 2027	2,500	98.19	58.0	96.7	60.0
USD 2028	3,000	97.73	58.0	91.4	63.5
USD 2029	2,125	98.25	58.0	107.0	54.2
USD 2030	1,400	98.72	58.0	99.9	58.1

Notes:

^a Recovery rates do not include the amounts of Arrear Bonds delivered as compensation for accrued interest.

^b The USD 2024 were issued under a 2014 indenture that only featured series-by-series CACs; all other bonds were governed by a 2016 indenture featuring enhanced CACs.

D Uniform Applicability in ICMA and Euro-Area CACs

This section provides a detailed summary of when the “Uniformly Applicable” condition is satisfied in the restructuring of multiple bond series under single-limb voting.

D.1 ICMA CACs

As stated in paragraph (c)-(v) of [ICMA \(2014\)](#), the “Uniformly Applicable” condition is satisfied if either (i) the eligible bondholders are offered the same new instrument or menu of new instruments, at the same terms (principal, accrued but unpaid interest, past due interest, modulo currency differences); or (ii) the eligible bonds are amended in such a way that the amended instruments have identical terms, except for currency differences.

D.2 Euro CACs

As stated in paragraph (1)-(k) of [EU \(2021\)](#), the “Uniformly Applicable” condition is satisfied if the eligible bonds have (i) their principal amount outstanding reduced by the same proportion; and/or (ii) their principal or interest repayment dates extended by the same period or the same proportion; and/or (iii) the ICMA requirements are satisfied (same menu of instruments or amendments that result in identical amended instruments).

E Supplementary Material for Section 6

E.1 Analytical Results in the Uniform Case

Under the assumption that G is uniform, market clearing in the threshold equilibrium that we consider requires that $\hat{r} = r_{\min} + \lambda_L R$, where $R \equiv r_{\max} - r_{\min}$ denotes the length of $\mathcal{R}$.

Reservation Value Distributions. The supports of the reservation value CDFs are

$$\mathcal{V}_L = [v_L(\hat{r}), v_L(r_{\min})] = \left[\frac{c_L}{\hat{r} + \delta_L + \chi}, \frac{c_L}{r_{\min} + \delta_L + \chi} \right] =: [\underline{w}_L, \bar{w}_L], \quad (\text{E.1})$$

$$\mathcal{V}_S = [v_S(r_{\max}), v_S(\hat{r})] = \left[\frac{c_S}{r_{\max} + \delta_S + \chi}, \frac{c_S}{\hat{r} + \delta_S + \chi} \right] =: [\underline{w}_S, \bar{w}_S]. \quad (\text{E.2})$$

Using the uniformity assumption, we can write for all $w \in \mathcal{V}_L$

$$\Pr(v_L(r) \leq w \mid r \leq \hat{r}) = \frac{G(\hat{r}) - G(c_L/w - (\delta_L + \chi))}{\lambda_L} = \frac{r_{\min} + \lambda_L R + \delta_L + \chi}{\lambda_L R} - \frac{c_L}{\lambda_L R} \frac{1}{w}.$$

Likewise, for $w \in \mathcal{V}_S$, we have

$$\Pr(v_S(r) \leq w \mid r \geq \hat{r}) = \frac{1 - G(c_S/w - (\delta_S + \chi))}{\lambda_S} = \frac{R + \delta_S + \chi + r_{\min}}{\lambda_S R} - \frac{c_S}{\lambda_S R} \frac{1}{w}.$$

Let us define

$$\begin{aligned} a_L &= \frac{\lambda_L r_{\max} + (1 - \lambda_L) r_{\min} + \delta_L + \chi}{\lambda_L (r_{\max} - r_{\min})}, & b_L &= \frac{c_L}{\lambda_L (r_{\max} - r_{\min})}, \\ a_S &= \frac{r_{\max} + \delta_S + \chi}{\lambda_S (r_{\max} - r_{\min})}, & b_S &= \frac{c_S}{\lambda_S (r_{\max} - r_{\min})}. \end{aligned} \quad (\text{E.3})$$

Using these notations, the CDFs for the two bonds can be written as

$$F_i(w) = \begin{cases} 0 & \text{if } w < \underline{w}_i, \\ a_i - b_i/w & \text{if } w \in \mathcal{V}_i, \\ 1 & \text{if } w > \bar{w}_i, \end{cases} \quad i \in \{S, L\}. \quad (\text{E.4})$$

and one can check that $\underline{w}_i = b_i/a_i$ and $\bar{w}_i = b_i/(a_i - 1)$.

Auxiliary Problem and Two-Limb Offer. The auxiliary problem is

$$\min_{w_S, w_L} \lambda_S w_S + \lambda_L w_L \quad \text{subject to} \quad \lambda_S F_S(w_S) + \lambda_L F_L(w_L) = \tau_2^a,$$

and we denote its solution by $\hat{\mathbf{w}} = (\hat{w}_S, \hat{w}_L)$. In particular, when $\hat{\mathbf{w}} \in \text{int}(\mathcal{V}_S) \times \text{int}(\mathcal{V}_L)$, the first-order optimality condition

$$f_L(w_L) = f_S(w_S), \quad \text{with } w_S = F_S^{-1} \left(\frac{\tau_2^a - \lambda_L F_L(w_L)}{\lambda_S} \right) \quad (\text{E.5})$$

implies that

$$\hat{w}_S = \frac{\lambda_S b_S}{\lambda_S a_S + \lambda_L a_L - \lambda_L \frac{b_L}{\hat{w}_L} - \tau_2^a} \quad \text{and} \quad \hat{w}_L = \frac{\lambda_S b_S \sqrt{\frac{b_L}{b_S}} + \lambda_L b_L}{\lambda_S a_S + \lambda_L a_L - \tau_2^a},$$

which, after substituting the expressions for the coefficients (a_i, b_i) from (E.3), yields

$$\begin{aligned} \hat{w}_S &= \frac{c_S \left(\sqrt{\frac{c_L}{c_S} \frac{\lambda_L}{1 - \lambda_L}} + 1 \right)}{(\lambda_L - \tau_2^a) R + r_{\max} + r_{\min} + \delta_S + \delta_L + 2\chi}, \\ \hat{w}_L &= \frac{c_L \left(\sqrt{\frac{c_S}{c_L} \frac{1 - \lambda_L}{\lambda_L}} + 1 \right)}{(\lambda_L - \tau_2^a) R + r_{\max} + r_{\min} + \delta_S + \delta_L + 2\chi}. \end{aligned}$$

As in Section 2.2, the optimal two-limb offer coincides with the auxiliary solution $\hat{\mathbf{w}}$ when the latter satisfies the per-series constraint for both bonds.

Single-Limb Offer. Let $F(w) = \lambda_S F_S(w) + \lambda_L F_L(w)$ denote the aggregate consent share associated with uniform offer w , with $F'(w) \geq 0$. The optimal single-limb offer is given by

$$u^* = \inf \{ w \geq 0 \mid F(w) = \tau_1 \}.$$

In particular, for parameter values such that

$$\lambda_S \in \left[\frac{t_S + \frac{c_S}{c_L} (R\tau_1 - t_L)}{R}, \frac{t_L + \frac{c_L}{c_S} (R\tau_1 - t_S)}{R} \right],$$

where $t_i = r_{\max} + \delta_i + \chi$, one can show that

$$u^* = \frac{\lambda_L b_L + \lambda_S b_S}{\lambda_L a_L + \lambda_S a_S - \tau_1} = \frac{c_L + c_S}{(\lambda_L - \tau_1)R + r_{\max} + r_{\min} + \delta_S + \delta_L + 2\chi} \in \mathcal{V}_S \cap \mathcal{V}_L.$$

Impact of Relative Bond Size on Restructuring Outcomes. In the extended setup with sorting, a change in relative face values affects the equilibrium restructuring outcome through two distinct channels: the pure size effect discussed in Section 2, and an *additional* channel driven by the endogenous change in reservation value distributions. Consider for instance an increase in the relative size λ_S of the short-term bond, keeping other parameter values fixed. As captured by (E.1)–(E.2), a higher share λ_S of the short-term bond (i.e., a lower λ_L), by reducing the threshold $\hat{r}$, raises the upper bound of $\mathcal{V}_S$ and the lower bound of $\mathcal{V}_L$, without affecting the other bounds: F_S becomes flatter and F_L steeper (see also (E.3)–(E.4)). The change in restructuring outcomes driven by this change in reservation value CDFs—keeping the weight of the short-term bond at its baseline value—constitutes a new “sorting channel” that operates on top of the pure size effect.

E.2 Model Calibration

We apply the model to the 2020 Argentine restructuring of Macri bonds, discussed in detail in Section 4.2 of the paper and in Appendix C.2. The application focuses on USD-denominated bonds, which accounted for 86% of the total principal amount of Macri bonds, to allow for a clean and transparent exercise. In view of the differences in exchange offers and market recovery rates, we sort these bonds into two groups according to their maturities: we take bonds with maturities up to 2036 to construct the “short-term” bond basket, and bonds with longer maturities to construct the “long-term” basket. To compute pre-restructuring price differentials, we take close prices from LSEG Workspace on March 5th, 2020, i.e., six months

before the exchange settlement date. We then compute the price of each basket by taking the average price across bonds, weighted by their principal amount outstanding. We also take weighted averages to compute durations and market recovery rates for each basket, where bond durations (given in Table C.2) are computed using a reference exit yield—as explained in Appendix C.2—and the market recovery rates for each Eligible Bond (given in Table C.9) are computed as the average exit price across New Bonds, weighted by the share of orders submitted for each New Bond.

E.3 Impact of CACs on Bond Valuations

The counterfactual exercise conducted in Section 6.2 illustrates how the design of CACs affects bond valuations, focusing on the equilibrium price differential (Δq^*) between the short- and long-term bonds. The specification of CACs affects the price differential in the trading stage through its impact on the equilibrium offers in the restructuring stage. Figure E.1 further illustrates how the impact of CACs on bond prices depends on the probability of default (η). Holding all other parameters fixed to their baseline calibration values, the left panel depicts the valuation by the marginal investor of short- and long-term bonds (in red and blue, respectively) under two-limb and single-limb CACs (with solid and dashed lines, respectively). Naturally, bond valuations are decreasing in the probability of default. Valuations under single-limb CACs are higher than under two limb due to the more generous recovery values extended by the government to both bonds in order to meet the higher aggregate voting threshold, and the valuation differential across CAC specifications is increasing in the likelihood of a restructuring. The right panel shows that the specification of CACs has a greater impact on the bond price differential (Δq^*) when the probability of a future restructuring is larger.

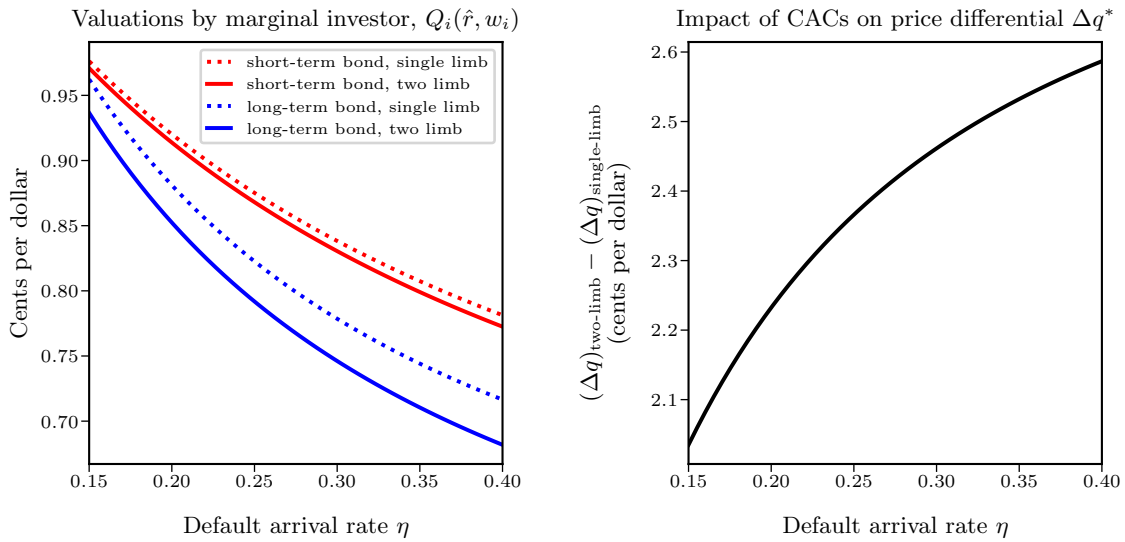


Figure E.1: Impact of CAC Type on Bond Valuations.